9 The Gravitational Field

Classical field theory is primarily a study of electromagnetic and gravitational fields. This chapter is an *introduction* to field theory and is limited to a few aspects of the gravitational field.

Perhaps the most thoroughly studied field is the electromagnetic field. You will learn about it in your E&M course. However, field theory is used to study many other physical phenomena. For example, field theory is used in fluid dynamics and in studies of elasticity, not to mention such important research areas in physics as the unified field theory (which hopes to unite gravity and electromagnetism) and quantum field theory. An interesting aspect of quantum field theory is that it associates particles with the fields. Relativistic quantum field theory has become an important tool in high energy physics. As you might imagine, field theory is quite complicated, both conceptually and mathematically. You may be relieved to know that in this chapter we will limit ourselves to a few fairly straightforward concepts.

An interesting and important aspect of field theory is the formation and propagation of waves. When you study the electromagnetic field you will spend a significant amount of time and mental energy learning about electromagnetic waves. These waves are of great practical importance, particularly because X-rays, visible light, radio waves, and infrared radiation are all electromagnetic waves. The gravitational field also gives rise to waves. Einstein's general theory of relativity predicts that gravity waves are generated by the acceleration of massive bodies. Thus, gravity waves are expected to be produced by supernova explosions, black hole mergers, and orbiting pulsars. The first direct detection of gravitational waves was made with a very large (and complex) system called LIGO.[2]

9.1 Newton's Law of Universal Gravitation

The concept of gravitation was put on a mathematical basis by Isaac Newton when he formulated the law of universal gravitation. He postulated that every particle in the universe attracts every other particle with a force that is proportional to the product of their masses and inversely

[1] To learn more about field theory and the closely related subject called potential theory, see L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields: Course of Theoretical Physics*, Vol. 2. Pergamon Press, 1975. A very old but very good book that was first published in 1929 is Oliver Dimon Kellogg, *Foundations of Potential Theory*, J. Springer, 1929 (available as a republication from Dover Publications).

[2] LIGO stands for Laser Interferometer Gravitational-Wave Observatory. On September 14, 2015, LIGO observed gravitational waves from the merger of two black holes. More details are found in the article by B. P. Abbott and many co-authors in *Physical Review Letters* (DOI:10.1103/PhysRevLett.116.061102), and in the news article by Adrian Cho in *Science*, February 11, 2016.

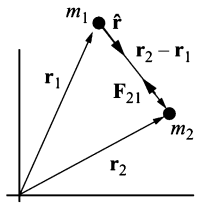


Figure 9.1 A body of mass m_1 is located at $\mathbf{r}_1$ and a body of mass m_2 is located at $\mathbf{r}_2$. The relative position of m_2 with respect to m_1 is $\mathbf{r}_2 - \mathbf{r}_1$.

proportional to the square of the distance between them. In equation form, Newton's law of universal gravitation states that the force on a particle of mass m_2 due to a particle of mass m_1 is

$$\mathbf{F}_{21} = -G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^2}\hat{\mathbf{r}}, \tag{9.1}$$

where G is a constant of proportionality, known as the universal gravitational constant and found experimentally[3] to be approximately equal to 6.67×10^{-11} Nm^2/kg^2.

As shown in Figure 9.1, the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are the position vectors of the two particles, and $\hat{\mathbf{r}}$ is a unit vector pointing from m_1 to m_2. This unit vector is given by

$$\hat{\mathbf{r}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|}.$$

Therefore, a slightly different way to express the law of gravitation is

$$\mathbf{F}_{21} = -Gm_1 m_2\frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3}. \tag{9.2}$$

Although this is more complicated, it avoids confusion as to the direction of the force. Recall that $\mathbf{F}_{21}$ is the force *on* m_2 due to m_1 and that the negative sign in the equation indicates that the force is attractive. By Newton's third law, the force on particle m_1 due to particle m_2 is

$$\mathbf{F}_{12} = -Gm_2 m_1\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}.$$

What is the force law for extended bodies? If the distance between two bodies is much larger than the dimensions of the bodies (as for astronomical objects), it is usually safe to treat the bodies as particles. The quantity $|\mathbf{r}_2 - \mathbf{r}_1|$ is the distance between the centers of the two bodies. If an extended body cannot be treated as a particle, you will need to determine the gravitational field of the extended body, as discussed below in Section 9.3. (For example, if a satellite is in a near-Earth orbit, the Earth cannot be considered a point mass.)

9.1.1 Universality of the Law of Gravitation

According to Newton's law of gravitation, every object (every atom) in universe is exerting a force on every other object in the universe. It is true that the force decreases as $1/r^2$ (inverse square force law) so the force we exert on the distant stars is entirely negligible. Nevertheless, a person on the Earth is exerting a tiny force on all the planets and stars and galaxies.[4]

[3] A recent measurement of G by Guglielmo Tino and collaborators at the University of Florence, Italy, yielded a value of $G = 6.671\,91(99) \times 10^{-11}$ m^3 kg^{-1} s^{-2}. See the article "Precision Measurement of Newtonian Gravitational Constant Using Cold Atoms," by G. Rosi *et al., Nature*, **510**, 518–521 (2015).

[4] The British physicist P. A. M. Dirac is reputed to have expressed this aspect of the law of gravitation with the poetic sentiment, "When you pluck a flower, you move a distant star."

9.1.2 Action at a Distance

Newton's law of universal gravitation is a prime example of the concept of "action at a distance" which means that body A exerts a force on B, no matter how far apart the bodies happen to be. Naturally, the question arises, "How can bodies that are separated by great distances exert forces on one another?". Newton stated, "Hypothesis non fingo," which means, "I make no hypothesis." This is a perfect example of physics telling us *how* nature works but making no assumptions (or hypotheses) as to *why* it works that way.

The biggest problem with the concept of action at a distance is that it implies that the force between two bodies is propagated instantaneously. That is, if body A is moved to another position, body B will instantaneously notice a change in the force acting on it, even if B is halfway across the universe. Clearly, this is impossible. We believe (but cannot yet prove) that the gravitational force is transmitted at the speed of light. The conceptual difficulties inherent in Newton's law of universal gravitation expressed in terms of action at a distance are effectively dealt with when one considers the gravitational *field*, rather than the gravitational *force*.

Exercise 9.1

Determine the force exerted by Mars on the Sun. What is the acceleration of Mars? What is the acceleration of the Sun? ■

Exercise 9.2

A body falls in the gravitational field of the Earth. Show that its acceleration is independent of its mass. ■

Exercise 9.3

A particle of mass m is located at (1,2), a particle of mass $2m$ is located at (3,4), and a particle of mass $3m$ is located at $(-2, -2)$. Determine the gravitational force acting on the particle of mass $2m$. Answer: $F = -Gm^2(0.24\hat{\mathbf{i}} + 0.25\hat{\mathbf{j}})$. ■

Exercise 9.4

Prove that the gravitational force is conservative by evaluating the curl of Newton's law of universal gravitation. ■

9.2 The Gravitational Field

By definition, *a field is a physical quantity that is defined at all points in some region of space.* For example, the temperature is a physical quantity and using a thermometer you could determine the temperature at every point in this room. This scalar field might be described by an equation $T = T(x, y, z)$, or by a table of values, or by a drawing showing the temperature isopleths of the room. Similarly, the water in a river has a velocity at every point. The region of space is the river and the physical quantity is velocity. This vector field might be represented graphically or perhaps by an equation for the velocity as a function of position. Suppose that somehow you could determine the gravitational force on a mass at all points near a planet. This would define the gravitational force field of the planet. Note that in all of these examples, the value of the field depends on position.

To define the gravitational field more explicitly, consider a region of space that is empty except for two particles. One particle has mass M and it is located at a fixed point in space. At some other point you place a particle of much smaller mass m. Assume you have a way to determine the force acting on the small mass, wherever it may happen to be. The small mass will be the "test body."

Now imagine you move this test body from one point to another and you measure the magnitude and the direction of the gravitational force acting on it at every location. If you represent the force by a vector, you will end up with a lot of force vectors all pointing towards M. Unfortunately, as indicated by Equation (9.1), these force vectors depend on the mass of the test body so they are not a good measure of the gravitational field of M. To avoid any dependence on the test body, you can define the gravitational field of M as the force *per unit test mass* at all points in space around M. Furthermore, because you don't want m to have any influence on the field, you take the limit as $m \to 0$.

These concepts are expressed more simply using mathematics. Let $\mathbf{F}$ be the *force* on m due to M and let $\mathbf{g}$ be the gravitational *field* of M. Then, by definition, if m is at $\mathbf{r}$ and M is at $\mathbf{r}'$,

$$\mathbf{g}(\mathbf{r}) = \lim_{m\to 0}\left(\frac{\mathbf{F}}{m}\right) = -GM\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}. \tag{9.3}$$

The notation $\mathbf{g} = \mathbf{g}(\mathbf{r})$ reminds us that $\mathbf{g}$ is a function of position.

The location of body M is called the "source point," and its position is denoted by $\mathbf{r}'$. The point specified by $\mathbf{r}$ is called the "field point." It is the place where the field is evaluated. See Figure 9.2. Note that the test mass does not enter into Equation (9.3), the expression for $\mathbf{g}(\mathbf{r})$.

Introductory physics books often place the origin of coordinates at the mass M. This gives a simpler but less descriptive expression for the field:

$$\mathbf{g}(\mathbf{r}) = -GM\frac{\mathbf{r}}{|\mathbf{r}|^3} = -\frac{GM}{r^2}\hat{\mathbf{r}}, \tag{9.4}$$

where $\mathbf{r}$ is the location of the field point.

Let me be very clear about the difference between the force and the field. The force $\mathbf{F}$ depends on the masses of both of the particles and their positions at some instant of time. The *field* $\mathbf{g}$ depends only on the mass of the source particle M and the relative position of the field point. The force $\mathbf{F}$ describes an interaction between two objects, but the field $\mathbf{g}$ is a property of a point in space.[5]

The point mass M is the *source* of the gravitational field. The field extends throughout all space. It is not easy to grasp the field concept; it is particularly difficult to describe exactly *what* is filling all of space. We know that if we place a test mass at any point in the field it will feel a

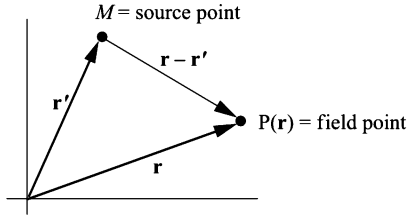


Figure 9.2 Particle M at the source point $\mathbf{r}'$ generates a gravitational field everywhere. The gravitational field at point $\mathrm{P}(\mathbf{r})$ (the "field point") is given by Equation (9.3).

[5] Sometimes, I like to change the definition of field slightly and state, "A field is a region of space in which some physical quantity is defined at every point." This focuses the mind on the *space* rather than on the physical quantity. It is not, however, the standard definition of field.

force, so we think of the mass M as producing *something* which exerts a force on any other mass in this space. The field is everywhere, but it exerts a force only when a material body is placed in the field.

At the surface of the Earth the gravitational field is approximately given by $\mathbf{g} = -9.8\ \text{m/s}^2\,\hat{\mathbf{k}}$. Here $\hat{\mathbf{k}}$ is a unit vector pointing upward at the surface. For points far above the surface, the field is approximately $\mathbf{g}(\mathbf{r}) = -(GM_E/r^2)\hat{\mathbf{r}}$, where the origin is at the center of the Earth and M_E is the mass of the Earth. (Do not confuse the field vector $\mathbf{g}(\mathbf{r})$ with the quantity g which is an abbreviation for the numerical value $9.8\ \text{m/s}^2$.)

The field concept is quite different from action at a distance. In action at a distance we consider the force one body exerts directly on another. When using the field concept, we think of the interaction of bodies M and m as a two-step process in which mass M generates a field and then mass m interacts with the field, rather than interacting with M directly. Thus the field approach *decouples* the sources from the test body used to determine the field. If two different source arrangements generate the same field at some point, a test body at that point will feel the same force.

It is tempting to consider the field to be nothing but a convenient way of expressing the force, but it turns out that the field has an existence of its own, independent of the force. A force is just a force, but a field has energy, momentum, and angular momentum as well as the ability to exert a force. A field can propagate through space as a wave and can exist independent of the source. (The field can persist even after the source has ceased to exist.)

Although the field approach seems more complicated because it introduces a new concept ("the field"), it does resolve a number of difficulties with the action at a distance concept. As mentioned, in the scenario of action at a distance, forces are transmitted instantaneously. In the field concept, the response of the field to changes in the source propagate at a finite speed. Because the source (M) generates the field, if the source is moved, the field will change. However, this change does not occur instantaneously at all points in space. For example, we believe that changes in the gravitational field are propagated at the speed of light.

Furthermore, the field approach leads to a conceptual framework for understanding how the force is transmitted. In quantum field theory the interaction between particles can be described as an interchange of "virtual" particles. The electromagnetic force, for example, is believed to be transported by virtual photons. This is often represented graphically by "Feynman diagrams." For example, the electromagnetic interaction between two electrons ("electron–electron scattering") is represented by the Feynman diagram of Figure 9.3. In the figure, the two electrons are represented by the straight lines and the wavy line represents a virtual photon where the word "virtual" is used because these photons are not (and cannot be) observed. The photon transmits the electromagnetic force. It is emitted by one of the electrons and absorbed by the other. Both particles are deflected from their original paths. Similarly, the gravitational force is assumed to be transmitted by undetected particles called "gravitons." Because we have our hands full with just Newton's theory of gravitation, we will not consider these advanced concepts in this book.

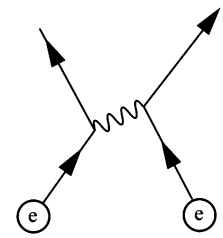


Figure 9.3 A Feynman diagram for electron–electron scattering. The wavy line represents the virtual photon.

However, if you are interested in delving deeper into quantum field theory, I recommend you read a very short but very profound book by Richard Feynman called *QED*.[6]

Exercise 9.5

Two particles of mass M are separated by a distance a. Determine the gravitational field at a distance z along the perpendicular bisector of the line joining the masses. (Remember that in Equation (9.3) the field point is at $\mathbf{r}$ and the source point is at $\mathbf{r}'$.) Answer: $\mathbf{g} = -2GMz\hat{\mathbf{z}}/\left[(a/2)^2 + z^2\right]^{3/2}$. ■

Exercise 9.6

A point mass M is located at the origin. Another point mass M is located at (0,4). Determine the gravitational field at (0,3) and at (3,0). Answer: Field at (3,0) is $\mathbf{g} = GM\left[-0.14\hat{\mathbf{i}} + 0.03\hat{\mathbf{j}}\right]$. ■

Exercise 9.7

Determine the gravitational field due to the Earth and the Sun at the position of the Moon: (a) at the time of a lunar eclipse, and (b) at the time of a solar eclipse. (c) Determine the force on the Moon at these two times. ■

9.3 The Gravitational Field of an Extended Body

The gravitational field produced by an extended body can be determined by treating the body as a collection of point masses and summing the fields due to each individual particle. (Gravitational fields obey the principle of superposition and add vectorially.) Although this is a perfectly *reasonable* thing to do, it is not a *practical* thing to do. So we assume that the material of the extended body is *continuous*. This assumption breaks down on the atomic level where there are huge differences in mass density between the small, heavy nuclei and the empty space that comprises most of the volume of atoms. However, a volume element that contains billions of atoms is still very small on the macroscopic scale and we are justified in thinking of it as containing a continuous distribution of matter with a well-defined density.

Figure 9.4 shows an infinitesimal volume element $(d\tau')$ in a continuous body. The mass contained in this volume element is $dm' = \rho d\tau'$, where $\rho = \rho(\mathbf{r}')$ is the density. (For a surface, the element of mass is σdA, for a line it is λds and for an extended body it is $\rho d\tau$, where σ, λ, ρ are the mass per unit area, per unit length, and per unit volume.) The density may vary from one

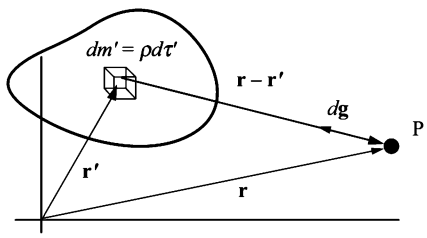


Figure 9.4 An extended body. The mass element $dm' = \rho d\tau'$ is located at $\mathbf{r}'$ (the source point) and it produces an infinitesimal gravitational field $d\mathbf{g}$ at point P located at $\mathbf{r}$ (the field point).

[6] Richard Feynman, *QED: The Strange Theory of Light and Matter*, Princeton University Press, Princeton, NJ, 1985.

part of the body to another so it is written as $\rho = \rho(\mathbf{r}')$ to remind us that ρ is a function of position.

In Figure 9.4 the point P is the field point. Note that the source point is at $\mathbf{r}'$ and the field point is at $\mathbf{r}$. The infinitesimal portion of the gravitational field at $\mathbf{r}$ due to an infinitesimal mass element dm' at $\mathbf{r}'$ is

$$d\mathbf{g}(\mathbf{r}) = -Gdm'\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}$$

or

$$d\mathbf{g}(\mathbf{r}) = -G\rho(\mathbf{r}')d\tau'\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3},$$

where $d\tau'$ is so small that $\rho(\mathbf{r}')d\tau'$ can be considered a particle, yet large enough to treat the mass as continuous. To obtain the field at P due to the whole body, we integrate over the body. That is,

$$\mathbf{g}(\mathbf{r}) = -G\int_{\text{body}}\rho(\mathbf{r}')\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}d\tau'. \tag{9.5}$$

For an extended body with sufficient symmetry, it is usually not too difficult to evaluate this integral. To find the field due to an arbitrarily shaped body, it is usually best to first determine the *potential* and then obtain the field, using the technique described in Section 9.4 below. Therefore, I will not emphasize the direct integration of Equation (9.5). Nevertheless, you should have some experience with this technique so I will work through two examples (Worked Examples 9.1 and 9.2). You can get some additional experience by working out the exercises below. It might help you to review the section on the electric field of a continuous charge distribution in your introductory physics textbook, as the techniques are exactly the same. It is usually easier to solve such problems by resolving the vector $d\mathbf{g}$ into its components before carrying out the integration.

Worked Example 9.1 A long thin rod of length L lies along the x-axis. The end points of the rod are at $x' = 0$ and $x' = L$. Assume the rod can be represented as a continuous mass distribution with linear density λ mass per unit length. Determine the gravitational field at the point $x = 4L$.

Solution In this problem, the mass element is $dm = \lambda dx'$, where the variable x' represents the distance from the origin to the mass element. The infinitesimal field at P due to this portion of the rod is

$$\begin{aligned} d\mathbf{g} &= -Gdm\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3} = Gdm\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3} \\ &= -G\lambda dx'\frac{(x-x')\hat{\mathbf{i}}}{|x-x'|^3} = -G\lambda\hat{\mathbf{i}}\frac{dx'}{(x-x')^2}. \end{aligned}$$

The field point is at $x = 4L$, so integrating we have

$$\mathbf{g} = -G\lambda\hat{\mathbf{i}}\int_{x'=0}^{x'=L}\frac{dx'}{(4L-x')^2} = -G\lambda\left[\frac{1}{4L-x'}\right]_0^L\hat{\mathbf{i}} = -\frac{1}{12L}\lambda G\hat{\mathbf{i}}.$$

Worked Example 9.2 Determine the gravitational field at a point P a distance $+z$ along the axis of a very thin disk of mass M, uniform density, and radius R. See Figure 9.5.

Solution Treat the disk as if it had zero thickness. Then the appropriate mass element is $\sigma\, dA$, where $\sigma = M/\pi R^2$. As indicated in Figure 9.5 the field at point P due to mass element dm' is the vector $d\mathbf{g}$ pointing towards dm'. The distance from the source point dm' to the field point is $|\mathbf{r} - \mathbf{r}'|$, denoted D for simplicity. Consider all the mass elements lying on a ring a distance r' from the axis. The $d\mathbf{g}$ vectors from P to these mass elements form a cone of vectors with vertex at P. The horizontal components of these vectors will cancel. The components along the axis will sum. The resultant field is given by

$$\mathbf{g} = \int (dg)\cos\theta\hat{\mathbf{k}} = -\int \frac{G\,dm'}{D^2}\cos\theta\hat{\mathbf{k}} = -G\int \frac{\sigma\, dA'}{D^2}\cos\theta\hat{\mathbf{k}}.$$

But $\cos\theta = z/D = z/(r'^2 + z^2)^{1/2}$, and $dA' = r'd\phi' dr'$, so

$$\mathbf{g} = -\sigma G\hat{\mathbf{k}}\int_{r'=0}^{R}\int_{\phi'=0}^{2\pi} z\frac{(dr')(r'd\phi')}{(r'^2+z^2)^{3/2}} = -\sigma G\hat{\mathbf{k}}(2\pi)z\int_0^R \frac{r'dr'}{(r'^2+z^2)^{3/2}}$$

$$= -\sigma G\hat{\mathbf{k}}(\pi)z\int_{u=z^2}^{u=R^2+z^2}\frac{du}{u^{3/2}} = -2\pi\sigma G\hat{\mathbf{k}}z\left[\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}}\right]$$

$$= -\frac{2GM}{R^2}\left(1 - \frac{z}{(z^2+R^2)^{1/2}}\right)\hat{\mathbf{k}}.$$

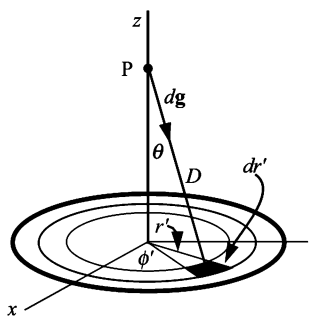


Figure 9.5 The gravitational field of a thin disk is determined by integration of the elementary field vector $d\mathbf{g}$.

Exercise 9.8

Determine the gravitational field at a point z on the axis of a ring of mass M and radius R. Answer: $-GMz/(z^2+R^2)^{3/2}$. ■

Exercise 9.9

Show that a disk can be treated as a collection of concentric rings. Write an integral expression for $\mathbf{g}$ using the result of Exercise 9.8 and show that it yields the expression obtained in Worked Example 9.2. ■

9.4 The Gravitational Potential

The gravitational *potential* is a scalar quantity related to the gravitational *field*, which is a vector quantity. It is usually easier to work with the potential than the field. If you know the potential you can always determine the field.

Recall the discussion in Section 5.3.3 proving that if $\mathbf{F}$ is a conservative force, then $\nabla \times \boldsymbol{F} = 0$. It is easy to show that the gravitational force is conservative. (See Exercise 9.4.) Using the fact that the curl of the gradient of any scalar function is zero, we conclude that since $\nabla \times \boldsymbol{F} = 0$, then $\mathbf{F}$ can be expressed as the gradient of some scalar function. That is, we can define the potential energy $V = V(\mathbf{r})$ such that

$$\mathbf{F} = -\nabla V.$$

Then

$$\int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{s} = -\int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{s} = -\int_1^2 dV = V_1 - V_2.$$

Thus, the difference in potential energy between two points is defined by

$$V_2 - V_1 = -\int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{s}.$$

In working with the gravitational potential energy, it is often convenient to define the zero of potential energy at infinity, so let $\mathbf{r}_1 = \infty$ and $V_1 = 0$. Writing $\mathbf{r}$ for $\mathbf{r}_2$ we have

$$V = -\int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{s}.$$

If $\mathbf{F}$ is the gravitational force acting on test body m located a distance r from point mass M, you can write

$$V(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} -\frac{GMm}{r^2}\hat{\mathbf{r}} \cdot d\mathbf{s} = -G\frac{Mm}{r}, \tag{9.6}$$

where $d\mathbf{s} = -\hat{\mathbf{r}}dr$. This is the gravitational potential energy of a system of two point masses.

When using the field approach, it is convenient to define a quantity called the *"gravitational potential"* which we will denote by the symbol Φ. The gravitational potential is obtained from the gravitational potential *energy* by dividing by m, the mass of the test body. The potential energy (V) of two interacting masses is an example of action at a distance and depends on the presence of two bodies. The potential (Φ) does not depend on the presence of a test body and is defined at every point in space. Thus, $\Phi = \Phi(\mathbf{r})$ is a field. The potential is related to the potential energy by

$$\Phi(\mathbf{r}) = \lim_{m \to 0} \frac{V(\mathbf{r})}{m},$$

where m refers to an infinitesimal test mass located at $\mathbf{r}$.

Dividing Equation (9.6) by m shows that the gravitational potential at $\mathbf{r}$ due to a particle of mass M located at the origin is

$$\Phi(\mathbf{r}) = -\frac{GM}{r}. \tag{9.7}$$

The potential of an extended body is obtained from

$$\Phi(\mathbf{r}) = -G\int_{\text{body}} \frac{\rho(\mathbf{r}')d\tau'}{|\mathbf{r}-\mathbf{r}'|}. \tag{9.8}$$

where $\mathbf{r}$ is the field point and $\mathbf{r}'$ is the location of the source point $dm = \rho(\mathbf{r}')d\tau'$.

Worked Example 9.3 Determine the gravitational potential at a point outside of a spherical shell of mass M and radius a. See Figure 9.6.

Solution The mass per unit area (σ) of the (uniform) shell is the total mass of the shell divided by its surface area, or

$$\sigma = \frac{M}{4\pi a^2}.$$

Selecting the mass element is more or less an "art" and you will become proficient at making a good choice after you have worked out a number of problems. For a mass shell, pick a small element of area on the surface whose mass is

$$dm = \sigma dA = \sigma a^2 \sin\theta d\theta d\phi.$$

Integrating over ϕ generates a ring on the surface of the shell, as indicated in the figure. (Integrating over ϕ is equivalent to letting the line $a\sin\theta$ rotate around $\mathbf{r}$. Note that all points on this ring are equidistant from the field point.) The width of the ring is $ad\theta$. The potential at P due to the entire shell is given by the double integral

$$\Phi = -G\int_{\text{surface}} \frac{\sigma dA}{|\mathbf{r}-\mathbf{r}'|} = -G\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} \frac{\sigma a^2 \sin\theta d\theta d\phi}{|\mathbf{r}-\mathbf{r}'|},$$

$$\Phi = -2\pi G\int_{\theta=0}^{\pi} \frac{\sigma a^2 \sin\theta d\theta}{|\mathbf{r}-\mathbf{r}'|},$$

where the factor 2π came from integrating over ϕ. Using the law of cosines, the figure shows that because $|\mathbf{r}'| = a$,

$$|\mathbf{r}-\mathbf{r}'| = R = \sqrt{a^2 + r^2 - 2ar\cos\theta},$$

and $R^2 = a^2 + r^2 - 2ar\cos\theta$. Therefore $2RdR = 2ar\sin\theta d\theta$ so $\sin\theta d\theta = \frac{R}{ar}dR$ and the integral takes the form

$$\Phi = -2\pi G\int_0^{\pi} \frac{\sigma a^2 \sin\theta d\theta}{R} = -2\pi G\sigma a^2 \int_{R=r-a}^{R=r+a} \frac{(R/ar)dR}{R}$$

$$= -G\sigma a^2(2\pi/ar)\int_{r-a}^{r+a} dR = \frac{-2\pi G\sigma a}{r}[(r+a)-(r-a)]$$

$$= -G\frac{4\pi a^2\sigma}{r} = -\frac{GM}{r},$$

where we used the fact that as θ goes from 0 to π, R ranges from $r-a$ to $r+a$.

Our result shows that for points *outside* the shell, the potential of a shell of mass M is the same as the potential of a point mass M at the origin. As a problem you can show that the *field* inside the shell is zero. (See Problem 9.15.)

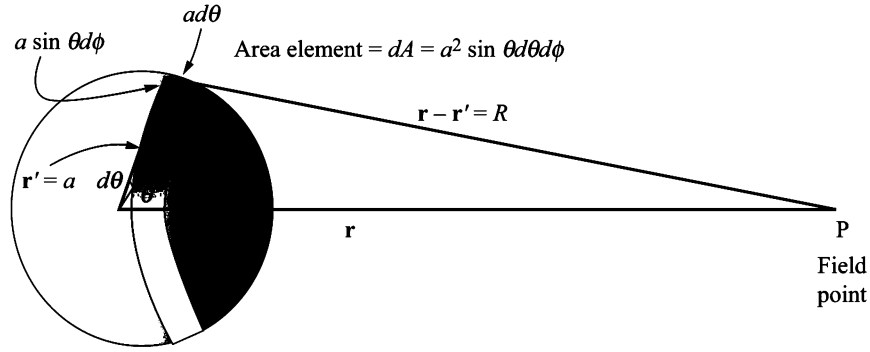


Figure 9.6 The potential at P due to a shell of radius a. The mass element dm is the mass of the element of area $dA = a^2 \sin\theta d\theta d\phi$. If σ is the surface mass density, $dm = \sigma dA$.

Exercise 9.10

Determine the potential on the axis of a disk of radius R, density ρ, and thickness t. Assume that $t \ll z$ so the thin disk approximation is still valid. Answer: $\Phi = -2\pi\rho t G[(z^2 + R^2)^{1/2} - z]$. ■

9.5 Field Lines and Equipotential Surfaces

We now illustrate how to represent the field and potential graphically.

Consider the gravitational field ($\mathbf{g}$) due to a point mass M, recall that $\mathbf{g}$ is the force per unit mass on an infinitesimal test body; see Equation (9.3). You could, in principle, transport the test body to every point in space around M and measure the force on it at each location. Then you could *represent* the field $\mathbf{g}$ in a number of different ways. For example, you could make up a large table giving the magnitude and direction of $\mathbf{g}$ at every point. Or you could represent $\mathbf{g}$ by an equation, such as Equation (9.4). Or you could represent $\mathbf{g}$ graphically. The graphical representation would show many arrows pointing toward M. The field at some given point might be represented by drawing a vector of appropriate length at that point, as illustrated by the sketch on the left in Figure 9.7. Note that the vectors are longer at points close to the mass and shorter at points further away. But this is not a very good way to represent the field. The sketch on the right in Figure 9.7 illustrates a much more convenient way. In this representation the lines give the direction of the force and the *areal density* of the lines at a point gives the magnitude of the force at that point. Imagine the lines are drawn in such a way that the number of lines crossing a unit area perpendicular to the field is numerically equal to the field at that location. If the field at some point has a magnitude of 15 N/kg, it could be represented by drawing 15 lines (per unit area) at that point.

The total number of lines passing through spherical surfaces centered on M will be the same regardless of the radius of the sphere. Because the area of a sphere increases as r^2, the density of lines per unit area falls off as $1/r^2$, just like the gravitational field.

The lines representing the field are traditionally called "lines of force," although most modern books refer to them with the better terminology of "field lines."

Another graphical representation of the field consists in sketching the surfaces on which the potential is constant. For a point mass, the equipotential surfaces are concentric spheres centered on the mass. As we will show in a moment, the field and the potential are related by $\mathbf{g} = -\nabla\Phi$,

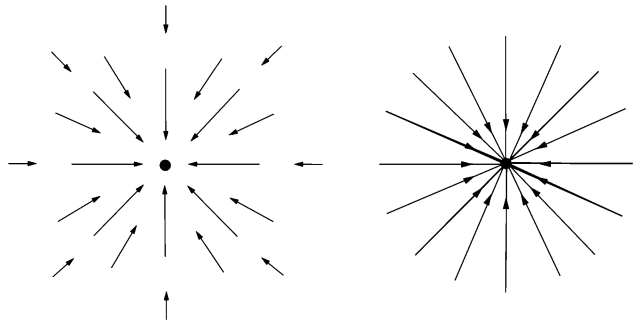


Figure 9.7 The force vectors are represented on the left. The strength of the force is represented by the length of the arrows. On the right, the field lines give the direction of the force and the density of field lines at any point is proportional to the strength of the field.

and you know from Chapter 5 that the gradient of Φ is perpendicular to the surfaces of constant Φ. Therefore, the two representations (field lines and equipotential surfaces) are equivalent.

9.6 The Newtonian Gravitational Field Equations

The relationship between potential energy and force is $\mathbf{F} = -\nabla V$, as shown in Equation (5.7). Dividing both sides of this equation by m yields the relation between the gravitational field $\mathbf{g}(\mathbf{r})$ and the gravitational potential $\Phi(\mathbf{r})$:

$$\mathbf{g} = -\nabla\Phi. \tag{9.9}$$

(In many books – particularly older books – the gravitational potential is defined as the *negative* of Φ and the relation equivalent to Equation (9.9) does not have the minus sign.)

The curl of a gradient is zero. Therefore,

$$\nabla \times \mathbf{g} = -\nabla \times (\nabla\Phi) = -\mathbf{curl}\ \mathbf{grad}\Phi = 0. \tag{9.10}$$

(I am sure you expected this result because you know that $\mathbf{g}$ is a conservative field.)

9.6.1 Gauss's Law

Having found the curl of $\mathbf{g}$, it is reasonable to go on and determine the divergence of $\mathbf{g}$. The process is a bit involved, so please follow carefully. Consider a mass point M. Draw a surface S around it, as illustrated in Figure 9.8. Let dS be an infinitesimal patch of the surface. Let $\hat{\mathbf{n}}$ be a unit vector perpendicular to dS and let $\mathbf{r}$ be the vector from M to dS.

Place the origin of a coordinate system at the point mass M. Then the gravitational field of M at the position of the patch of surface dS can be written as

$$\mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}},$$

so

$$\mathbf{g} \cdot \hat{\mathbf{n}} = -\frac{GM}{r^2}\hat{\mathbf{r}} \cdot \hat{\mathbf{n}} = -\frac{GM}{r^2}\cos\theta,$$

where θ is the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{n}}$ (see Figure 9.8).

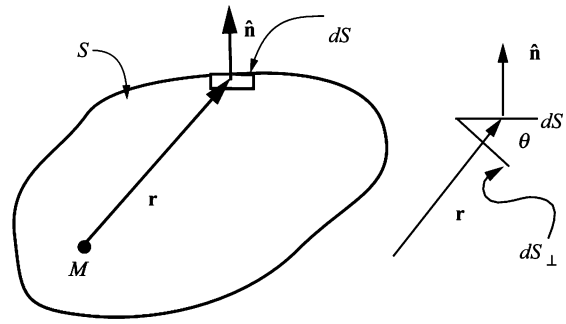


Figure 9.8 A closed surface S encloses a point mass M. The unit vector $\hat{\mathbf{n}}$ is perpendicular to the surface at the location of dS, an infinitesimal portion of the surface. Let $dS_\perp$ be the projection of dS perpendicular to $\mathbf{r}$. Note that $\mathbf{dS}\cdot\hat{\mathbf{r}} = dS\hat{\mathbf{n}}\cdot\hat{\mathbf{r}} = dS\cos\theta = dS_\perp$. The sketch on the right views the areas dS and $dS_\perp$ edge-on. The angle θ is the angle between the radial vector and $\hat{\mathbf{n}}$, or equivalently, between dS and $dS_\perp$.

Note that the solid angle[7] $d\Omega$ subtended by dS at M is

$$d\Omega = \frac{dS_\perp}{r^2} = \frac{dS\cos\theta}{r^2}, \tag{9.11}$$

where $dS_\perp$ is the perpendicular projection of dS on $\mathbf{r}$. Next consider the surface integral

$$\begin{aligned} I &= \oint_S \mathbf{g}\cdot\hat{\mathbf{n}}dS = -\oint_S \frac{GM}{r^2}\cos\theta dS \\ &= -GM\oint_S \frac{dS\cos\theta}{r^2} = -GM\oint_S d\Omega \\ &= -4\pi GM. \end{aligned}$$

This result is very interesting. It tells us that the surface integral I is proportional to the mass M inside the surface. This fact, in itself, is not too surprising. However, the result also tells us that *it does not matter where the mass is*, as long as it is inside the closed surface. If there are several masses $(m_1, m_2, \ldots)$ inside S, then, by the principle of superposition,

$$\oint_S \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -\sum 4\pi G m_i = -4\pi G\sum m_i = -4\pi G M_{\text{enc}}, \tag{9.12}$$

where M_{enc} is the total mass enclosed by S. The equation

$$\oint_S \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -4\pi G M_{\text{enc}}$$

is called Gauss's law. It is particularly useful in the study of electrostatics.

Assume mass is continuously distributed in a volume V bounded by a closed surface S. The element of mass is $\rho d\tau$, so the mass enclosed is $M_{\text{enc}} = \int_V \rho d\tau$. Therefore,

$$\oint_S \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -\int_V 4\pi G\rho d\tau. \tag{9.13}$$

[7] The solid angle is the three-dimensional angle formed at the vertex of a cone. Let point O be at the origin and a surface of area dA be a distance r from the origin. Then, if dA is perpendicular to r, the solid angle subtended by dA at O is $d\Omega = dA/r^2$. If dA is not perpendicular to r, $d\Omega = (dA\cos\theta)/r^2 = dA_\perp/r^2$. The units of solid angle are *steradians*. The solid angle subtended by a spherical shell at a point inside the shell is 4π steradians.

You may recall from your vector analysis course that Gauss's divergence theorem states that for any vector function **A**,

$$\oint_S \mathbf{A}\cdot\hat{\mathbf{n}}\,dS = \int_V \nabla\cdot\boldsymbol{A}\,d\tau. \tag{9.14}$$

Therefore, the left-hand side of Equation (9.13) can be expressed as $\int_V \nabla\cdot\mathbf{g}d\tau$, and consequently

$$\int_V \nabla\cdot\mathbf{g}\,d\tau = -\int_V 4\pi G\rho d\tau,$$

or

$$\nabla\cdot\mathbf{g} = -4\pi G\rho. \tag{9.15}$$

Thus, the divergence of **g** is proportional to ρ, the mass density. An important theorem called the Helmholtz theorem states that for any vector field, if one knows the divergence and the curl of the field, then one can determine the field itself. For this reason, $\nabla\cdot\mathbf{g}$ and $\nabla\times\mathbf{g}$ are called the "sources" of **g**. Equations (9.10) and (9.15) indicate that the source of the gravitational field is the mass density ρ. In other words, masses generate gravitational fields.

Worked Example 9.4 Use Gauss's law to determine the gravitational field outside of an infinitely long cylinder of radius a with constant linear mass density λ.

Solution To solve, we construct a *Gaussian surface* which is a closed surface that has the same symmetry as the mass distribution. We select this surface so that on the surface $\mathbf{g}\cdot\hat{\mathbf{n}}$ is either constant or zero. The field point lies at an arbitrary point on this surface. For this problem the appropriate Gaussian surface is a cylinder of length L and radius r, where $r > a$. See Figure 9.9.

On the curved face of this "tin can" surface, the field is directed towards the axis of the cylinder, and **g** is constant. On the two end faces, **g** and $\hat{\mathbf{n}}$ are perpendicular. Therefore, Gauss's law gives

$$\oint_S \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = \int_{\text{side}} \mathbf{g}\cdot\hat{\mathbf{n}}\,dS + \int_{\text{ends}} \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -4\pi G M_{\text{enc}}.$$

Now

$$\int_{\text{side}} \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -g\int dS = -g(2\pi r)L,$$

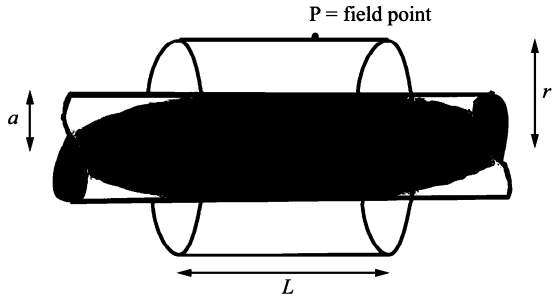


Figure 9.9 A Gaussian surface. The outer cylinder is the Gaussian surface appropriate for determining the field of the cylindrical mass. Note that the field point is on the Gaussian surface.

and

$$\int_{\text{ends}} \mathbf{g} \cdot \hat{\mathbf{n}}\, dS = 0.$$

The mass enclosed by the Gaussian surface is

$$M_{\text{enc}} = \lambda L.$$

Therefore,

$$g = -\frac{4\pi G\lambda L}{2\pi r L} = -\frac{2G\lambda}{r}.$$

By symmetry, this is pointing toward the axis of the cylinder so

$$\mathbf{g} = -\frac{2G\lambda}{r}\hat{\boldsymbol{\rho}},$$

where $\hat{\boldsymbol{\rho}}$ is the unit vector in cylindrical coordinates.

Exercise 9.11

Show that the solid angle subtended by a spherical shell at any interior point is 4π steradians. ■

Exercise 9.12

Use Gauss's law to determine the gravitational field of a point mass. ■

Exercise 9.13

Maxwell's equations for the electric field (**E**) and the magnetic field (**B**) can be written

$$\nabla \cdot \boldsymbol{E} = \rho/\epsilon_0 \quad \nabla \times \boldsymbol{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \boldsymbol{B} = 0 \qquad \nabla \times \boldsymbol{B} = \mu_0 \mathbf{J} + \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t},$$

where ρ is the charge density and J is the current density. The quantities ϵ_0 and μ_0 are constants. What are the sources of the electric field? What are the sources of the magnetic field? ■

9.7 The Equations of Poisson and Laplace

Recall that $\mathbf{g} = -\nabla\Phi$. Taking the divergence of both sides,

$$\nabla \cdot \mathbf{g} = -\nabla \cdot \nabla\, \Phi = -\nabla^2 \Phi.$$

Using the relation $\nabla \cdot \mathbf{g} = -4\pi G\rho$ leads to

$$\nabla^2 \Phi = 4\pi G\rho. \tag{9.16}$$

This is called "Poisson's equation."

Poisson's equation is a second-order partial differential equation whose solution depends on the boundary conditions. In a region of space where there is no mass, the density ρ is zero, and Poisson's equation reduces to

$$\nabla^2\Phi = 0. \tag{9.17}$$

This very important special case is called "Laplace's equation."

You will use Laplace's equation to determine the field in a region of space where $\rho = 0$, for example in empty space outside of a star. If, however, you want to determine Φ *inside* the star then you solve Poisson's equation.

Since Poisson's equation is an inhomogeneous differential equation, its solution can be expressed as the sum of two parts.

- The general solution of the homogeneous differential equation (Laplace's equation).
- A particular solution of the inhomogeneous differential equation (Poisson's equation).

In your undergraduate course in electromagnetic theory you will learn techniques for solving the Laplace equation under a variety of different conditions. In your graduate course in electromagnetic theory you will learn a number of methods for solving the Poisson equation.

Poisson's equation, $\nabla^2\Phi = 4\pi G\rho$ is the "field equation" of the Newtonian gravitational field. If you solve for Φ you know the potential everywhere and you can determine the gravitational field $\mathbf{g}$ everywhere.

9.8 Einstein's Theory of Gravitation (Optional)

We cannot leave this chapter on the gravitational field without a few words about Einstein's theory of gravitation, which is referred to as "The General Theory of Relativity." This theory supersedes Newton's theory of gravitation and is one of the greatest achievements in physics.

An example that illustrates the great accuracy of Einstein's theory is the precession of the orbit of Mercury. Astronomers had known for many years that the semimajor axis of the orbit of Mercury precesses at the extremely precisely measured rate of 5601 seconds of arc per century. Newton's theory, including the effects of the other planets which perturb the motion of Mercury, yielded a precession rate of 5558 seconds of arc per century. The discrepancy (43 seconds per century) was explained by Einstein's theory. Note that we are talking about a difference of less than one degree per century! There are numerous other examples of situations in which Einstein's theory gives the correct value whereas Newton's theory is slightly off.

In that case, you might ask, why do we bother studying Newton's theory? The answer is that calculations in Einstein's theory are extremely complicated, and that Newton's theory gives results that are extremely close to the correct values.

This section will give you a rough idea of Einstein's theory.[8] You will probably not be using the theory of general relativity unless you decide to specialize in cosmology – the study of the structure of the universe. Nevertheless, as a physicist, you should have some idea about the theory.

You may be aware that in 1905 Einstein developed the theory called the "Special Theory of Relativity." (We will consider this theory in Chapter 20.) The special theory deals with the relationship between two inertial reference frames. That is, it considers two reference frames that are moving at *constant* velocities. The laws of physics are expected to be the same in the two reference frames. In other words, a transformation from one reference frame to the other should

[8] A very readable book on the topic is *Relativity, Gravitation and Cosmology* by Robert J. A. Lambourne, Cambridge University Press, 2010.

leave physical equations and relationships unchanged. The theory was extremely successful and was particularly applicable to electromagnetism. However, Einstein was unsatisfied with one aspect of the special theory, and that was that Newton's law of universal gravitation did not transform as expected. Furthermore, Einstein was interested in developing a theory that would be applicable to *accelerated* (that is, noninertial) reference frames. In 1915 he published the "general" theory of relativity that solved both of these problems. However, to do so, he had to delve deeply into very advanced concepts in geometry.

Einstein's theories required understanding how to transform from one reference frame to another, a procedure that has much in common with transforming from one coordinate system to another. Recall that in Section 5.3.4 we introduced the concept of the "metric tensor" that allowed us to transform from Cartesian coordinates to spherical or cylindrical coordinates. Letting h_{ij}^2 represent the elements (or components) of the metric tensor, we wrote the differential (infinitesimal) displacement – see Equation (5.8) – as

$$ds^2 = \sum_{i,j} h_{ij}^2 dq_i dq_j.$$

In Sections 5.3.5 and 5.3.6 we used this relation to obtain expressions for the volume element and for del in cylindrical and spherical coordinates.

In special relativity, Einstein used a four-dimensional coordinate system with x, y, z being the usual Cartesian space coordinates and the fourth coordinate being ct, where c is the constant speed of light. Note that ct has the dimensions of distance, but it depends only on t. Thus, time is the fourth dimension. Including an extra dimension requires a somewhat more complicated expression for the differential displacement, namely:

$$ds^2 = \sum_{\mu,\nu=0}^{3} \eta_{\mu\nu} dx^\mu dx^\nu.$$

There are two things to note about this expression. First of all it is customary to use upper and lower indices which are called "contravariant" and "covariant." The difference need not concern us at present, but it is important to appreciate that dx^μ does *not* mean dx raised to the power μ! It is just a way to denote the component. Secondly, note that the indices μ and ν range from 0 to 3. Here 1, 2, 3 represent the spatial coordinates, dx, dy, dz and zero denotes the "time" coordinate cdt. For example, dx^3 represents dz and dx^0 represents cdt. The quantity ds is the infinitesimal "line element" in four-dimensional "space-time." For constant-velocity reference frames, the metric $\eta_{\mu\nu}$ is quite simple. It is called the "Minkowski metric"[9] and, in matrix form, it can be written as

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Minkowski space-time is described as "flat" because Euclidean geometry is valid in this space. This means, for example, that parallel lines never meet and that the angles in a triangle add up to 180°. The geometry of the surface of a sphere is non-Euclidean. Parallel lines that start out perpendicular to the equator meet at the pole. The angles of a triangle do not sum to 180°. In three-dimensional space, the line element ds on a flat surface is

[9] This metric was developed by Hermann Minkowski, who was one of Einstein's mathematics professors. Minkowski is credited for having developed the concept of space-time.

$$ds^2 = dx^2 + dy^2 + dz^2,$$

but on the surface of a sphere of radius R,

$$ds^2 = R^2 d\theta^2 + R^2 \sin^2\theta d\phi^2.$$

Not all surfaces that we might think of as curved are curved in the mathematical sense. For example, the curved side of a cylinder is mathematically flat because (as you can easily show) parallel lines never meet, triangles have 180°, etc.

Bernhard Riemann[10] used the concept of line element to generalize geometry. In Riemann's formulation, the line element in four-dimensional space-time is

$$ds^2 = \sum_{\mu,\nu=0}^{3} g_{\mu\nu} dx^\mu dx^\nu$$

In a flat region of space-time, the metric reduces to the Minkowski metric.

Einstein formulated the laws of physics in terms of tensors that reduce to the well-known forms of classical physics in the appropriate limit.

The basic relationship in Newtonian gravity is the Poisson equation:

$$\nabla^2\Phi = 4\pi G\rho.$$

Einstein wanted to replace this expression with an equivalent tensor equation. In Newtonian mechanics, mass is a conserved quantity. But according to relativity theory, mass and energy are interchangeable quantities. (Recall the famous equation $E = mc^2$.) Einstein decided to replace the mass density ρ with the energy–momentum tensor $T^{\mu\nu}$.[11] This is a 4×4 symmetric tensor which has 16 components, although only 10 of them are independent. These components are:

T^{00} = local energy density (including mass energy),
T^{0i} = momentum density in i direction times the speed of light,
T^{ij} = rate of flow per unit area of ith component of momentum perpendicular to the j direction $(i, j > 0)$.

In Einstein's theory, the energy–momentum tensor $T^{\mu\nu}$ multiplied by a constant of proportionality replaces $4\pi G\rho$ on the right-hand side of Poisson's equation.

Next Einstein needed to replace $\nabla^2\Phi$, the left-hand side of Poisson's equation, with a tensor. Note that $\nabla^2\Phi$ is a quantity that depends on the geometry of the space being considered. From Riemannian geometry the curvature of space-time can be expressed as a tensor which is denoted $G^{\mu\nu}$. Thus Einstein expressed the Poisson equation as $G^{\mu\nu} \propto T^{\mu\nu}$. It turns out that the proportionality constant is $8\pi G/c^2$ where G is the universal gravitational constant in Newton's law of gravity.[12] Consequently,

$$G^{\mu\nu} = -\frac{8\pi G}{c^2} T^{\mu\nu}. \tag{9.18}$$

This is, essentially, the Einstein expression for the gravitational field. It is a set of 10 coupled second-order differential equations for the metric $g^{\mu\nu}$ and is referred to as "Einstein's field

[10] Bernhard Riemann (1826–1866), who was a student of Carl Friedrich Gauss, is considered one of the world's greatest mathematicians. Among other achievements, he developed the geometry of curved surfaces.

[11] As you will learn in your E&M course, the basic laws of electricity and magnetism (Maxwell's equations) can be expressed in tensor form. (See, for example, Chapter 12 of *Introduction to Electrodynamics* by David. J. Griffiths, 4th ed., Cambridge University Press, 2017).

[12] G and $G^{\mu\nu}$ are two completely different things. Unfortunately, traditionally they are represented by the same letter. G is a scalar constant and $G^{\mu\nu}$ is a tensor describing the geometry of space-time and is a function of the $g^{\mu\nu}$.

equations." An interesting sidelight is the fact that when Einstein studied the implications of this set of equations, he discovered that they implied that space-time was increasing. That is, the equations predicted that the universe is expanding. At that time, astronomers were convinced that the universe was in a steady state, neither expanding nor collapsing. Bothered by this "fact," Einstein added a term to the equation to cancel out the expansion, thus

$$G^{\mu\nu} = -\frac{8\pi G}{c^2}T^{\mu\nu} - \Lambda g^{\mu\nu}.$$

The additional term is known as the "cosmological constant."

Then, in 1929, the astronomer Edwin Hubble discovered that distant galaxies are receding from us, and determined that the universe is, indeed, expanding.

Einstein called the introduction of the cosmological constant the "biggest blunder" of his life.

For many years, the cosmological constant was not included in Einstein's equations, but then in 1998 astronomers suggested that the universe is not only expanding, but this expansion is actually accelerating! Consequently, the cosmological constant has been reintroduced into Einstein's field equations to account for this increase in the rate of expansion. The present value of Λ is about $10^{-52}/\text{m}^2$.

Now that we have expressed Einstein's equations as an alternative to the Poisson equation, let us consider some of the consequences of this alternative approach to gravitation.

Perhaps the most fundamental question to be answered is what the theory predicts as to the effect on a particle (planet?) of mass m. How will a mass particle move in curved space-time? Recall our discussion of the calculus of variations in Chapter 4 in which we showed that the shortest distance between two points in a plane is a straight line and the shortest distance between two points on the surface of a sphere is a section of a great circle (or geodesic). It is an interesting consequence of general relativity that a particle in a gravitational field will move along a geodesic.

Furthermore, the theory predicts that a massive body will distort space-time, changing the curvature of space in its vicinity and thus affecting the geodesics.

The motion of a particle in curved space-time is often described in terms of the analogy of a bowling ball placed on a soft mattress, thus deforming the surface of the mattress. A marble rolling along the mattress will roll into the depression, as if attracted to the bowling ball. See Figure 9.10.

John Wheeler summarized the rules of general relativity in the following way:

Matter tells space how to curve.
Space tells matter how to move.

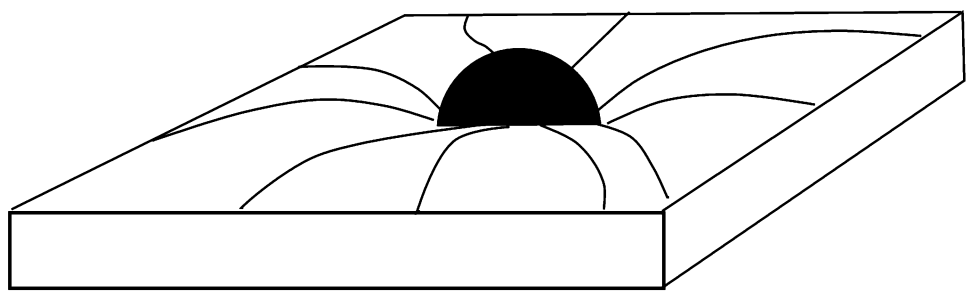


Figure 9.10 A bowling ball on a mattress as an analogue to curved space-time.

The theory of general relativity is basically a four-dimensional field theory of gravitation. It has been submitted to numerous experimental tests and has been proved correct over and over. It is unfortunate that it is so mathematically complex that it is only accessible to a small community of physicists, astronomers, and mathematicians.

9.9 Summary

A *field* is defined as a physical quantity whose value can be determined at every point in some given region of space. The gravitational field $\mathbf{g}(\mathbf{r})$ due to a particle of mass M located at point $\mathbf{r}'$ can be defined in terms of the force it exerts on an infinitesimal point mass m located at $\mathbf{r}$,

$$\mathbf{g}(\mathbf{r}) = \lim_{m\to 0}\left(\frac{\mathbf{F}}{m}\right) = -GM\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}.$$

The field produced by an extended body can be determined by evaluating the expression

$$\mathbf{g}(\mathbf{r}) = -G\int_{\text{body}} \rho(\mathbf{r}')\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}d\tau'.$$

To evaluate the integral it is usually necessary to resolve the vectors into components, using the symmetry of the problem as a guide.

The gravitational *potential* $\Phi(\mathbf{r})$ is given by

$$\Phi(\mathbf{r}) = -G\int_{\text{body}} \frac{\rho(\mathbf{r}')d\tau'}{|\mathbf{r}-\mathbf{r}'|}.$$

A field can be represented graphically by drawing the field lines (as illustrated in Figure 9.7) or by drawing equipotential surfaces.

The Helmholtz theorem tells us that we can determine a vector field if we know its "sources" defined as the gradient and curl of the field. For the gravitational field these are

$$\nabla\times\mathbf{g} = 0,$$
$$\nabla\cdot\mathbf{g} = -4\pi G\rho.$$

The first of these relations indicates that the gravitational field is conservative, thus allowing us to define a potential. The second equation is Gauss's law and in integral form it is expressed as

$$\oint_S \mathbf{g}\cdot\hat{\mathbf{n}}\,dS = -4\pi G M_{\text{enc}},$$

where M_{enc} is the mass enclosed by the surface S. In terms of the potential, this leads to

$$\nabla^2\Phi = 4\pi G\rho,$$

which is called *Poisson's equation*. In regions of space where $\rho = 0$ this reduces to *Laplace's equation*

$$\nabla^2\Phi = 0.$$

Finally, although Newton's theory of gravitation has been superseded by Einstein's theory of general relativity, we still use Newtonian mechanics for most practical problems because it is much easier to apply and for most situations the differences are miniscule.

9.10 Problems

Problem 9.1 Three particles of masses 1 kg, 2 kg, and 3 kg are located at (1,1), (2,0) and (3,3), respectively. They attract one another gravitationally. Determine the vector force acting on the 1 kg mass.

Problem 9.2 Assume the Earth's orbit is a circle of radius 1.496×10^{11} m. It orbits the Sun once a year. Using these facts, determine the mass of the Sun.

Problem 9.3 Suppose a person stood on a scale at the South Pole and the scale read 60 kg. What would it read at the equator? (Assume a perfectly spherical, homogeneous Earth.)

Problem 9.4 The Earth is suddenly brought to a standstill. Evaluate the time required for it to collide with the Sun. You can assume the Sun does not move and that the center of mass of the system is at the center of the Sun. How good is this approximation?

Problem 9.5 A neutron star has a mass of 10^{30} kg and a radius of 5 km. A body is dropped from a height of 20 cm above the surface. Determine the speed of the body when it hits the surface.

Problem 9.6 A neutron star is very small and very dense and rotates very rapidly. Assume the density of a particular neutron star is 5×10^{17} kg/m^3 and its mass is 5×10^{30} kg.

(a) Determine the gravitational force on a particle of mass 1 kg on the surface of the star.

(b) Assume the particle is on the equator of the star. Determine the maximum rotational velocity of the star if the gravitational force is sufficient to keep the particle on the surface.

Problem 9.7 Determine the period of a surface-skimming satellite in a circular orbit about a uniform, perfectly spherical planet of radius R and density ρ. Determine the period of such a satellite about a different planet which has radius $2R$ but the same density as the first. Explain your result.

Problem 9.8 (a) Determine the gravitational field at an arbitrary point above an infinite, thin, flat sheet having mass per unit area σ. (Hint: Use cylindrical coordinates.)

(b) A spherical shell of mass M and radius a is placed a distance b above the infinite flat plane ($b > a$). Determine the force on the sheet due to the shell.

Problem 9.9 Two concentric spherical shells of radii a and b have masses m_a and m_b. Assume $b > a$. What is the pressure (force per unit area) on the outer sphere?

Problem 9.10 Consider an infinitely long, straight string whose linear mass density is λ (mass per unit length). By direct integration, determine the gravitational field a distance r from the string.

Problem 9.11 By direct integration, find the field at a point on the axis of symmetry of a cylinder of length L, radius R and uniform density ρ. (Hint: Let the mass element be an infinitesimally thin disk and use the result of Worked Example 9.2.)

Problem 9.12 Consider an infinite string with linear mass density λ. A particle of mass m is a distance d from the string. Determine the force on the particle.

Problem 9.13 Obtain an expression for the potential at a distance z above the center of a circular disk of radius a and mass M.

Problem 9.14 Find the potential at an external point on the axis of a homogeneous right-circular cylinder which has mass M, length L, and radius a. The field point is a distance $z > L/2$ from the center of the cylinder. (Use the fact that a cylinder can be considered a collection of coaxial disks. The potential of a disk of mass dm is $d\Phi_{disk} = \frac{2Gdm}{a^2}[t - \sqrt{a^2 + t^2}]$, where t is the distance to the field point.)

Problem 9.15 Determine the potential *inside* a spherical shell of mass M. Use the technique of our analysis for the potential at an external point. Explain why your result predicts that there is no force on an object on the inside of a mass shell.

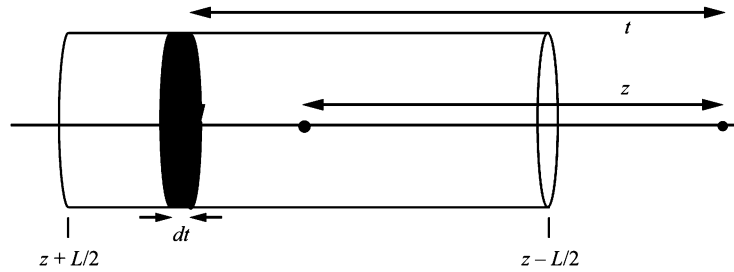


Figure 9.11 A homogeneous cylinder. Determine potential on axis. (Problem 9.14.)

Problem 9.16 Out in interstellar space there is a large spherical gas cloud of constant density ρ and radius a. Determine the gravitational potential at points inside the cloud.

Problem 9.17 Consider a planet of mass M and radius R. Assume the planet is spherical and has a constant density. By direct integration determine the gravitational field and the gravitational potential at all points inside and outside the planet. (Assume the potential is zero at infinity and there are no other bodies in the universe.)

Problem 9.18 (a) Using Gauss's law, determine the gravitational field a distance a from a flat infinite sheet of uniform mass density σ.

(b) Using Gauss's law, determine the field outside a spherical shell of radius a having the same mass density as the infinite plane.

(c) Show that just outside the shell, the field is twice that of the infinite plane. Interpret your result.

Problem 9.19 Suppose an intergalactic gas cloud were found to have a density given by

$$\rho = \frac{Mb^2}{2\pi r(r^2 + b^2)^2}.$$

Here M and b are constants related to the size and mass of the cloud. Note that the density extends out to infinity, but the density is infinitely small for very large values of r. Determine the gravitational field and the gravitational potential as functions of r.

Problem 9.20 Using Gauss's law, determine the gravitational potential inside and outside of a constant density sphere of radius R and mass M. Plot Φ vs. r.

Problem 9.21 Use Gauss's law to determine the field inside and outside of: (a) a sphere of uniform mass density, (b) a hollow spherical shell. Let the mass and radius be M and R in both cases.

Problem 9.22 Use Gauss's law to determine the field inside and outside an infinite cylinder of radius R and uniform mass density ρ. Express your answer in terms of r, the distance from the axis.

Problem 9.23 A sphere of radius R and constant mass density ρ has a spherical cavity of radius r, where $r = R/2$. The center of the cavity is a distance $R/4$ from the center of the sphere. A particle of mass m is located at an outside point, a distance z from the center of the larger sphere and along the line going through the centers of the sphere and the cavity. Determine the force on the particle. (Note: $z > R$.)

Problem 9.24 (a) Determine the gravitational potential at a point on the axis of a ring of mass of radius a and mass m. (b) Determine the potential for this ring at an off-axis point located a distance r from the center of the ring and making an angle θ with the axis. You may assume $r \gg a$ and you can disregard any terms of order $(a/r)^3$ or smaller. (Hint: The angle γ between two lines whose directions are specified by θ_1, ϕ_1 and θ_2, ϕ_2 is given by $\cos\gamma = \cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2)$.)

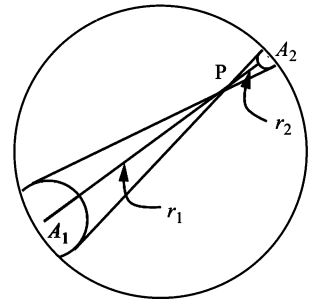


Figure 9.12 A spherical mass shell and two cones with common vertex at P.

Problem 9.25 Suppose a tunnel were dug through the Earth along a diameter. Show that an object dropped into this tunnel would undergo simple harmonic motion. Determine the period of the motion. (Air resistance is, of course, neglected.)

Problem 9.26 A particle is at an arbitrary point P inside a mass shell. (a) Draw the line through P so that it passes through the center of the shell (that is, the line is a diameter). Construct a "double cone" with vertices at P as shown in Figure 9.12. Show that the resultant gravitational force on the particle due to the two surface elements A_1 and A_2 is zero. (b) Generalize to any line through P and show that there is no net force acting on the particle at any interior point. (This is the way Newton showed that there was no force on a particle inside a shell.)

Problem 9.27 An infinite mass plane gives rise to a constant gravitational field, and a sphere of mass M gives rise to the gravitational field given by Newton's law of universal gravitation. Consider an object that looks like Saturn, but instead of rings, the equatorial plane of the sphere consists of a uniform mass plane of infinite extent and mass density σ (per unit area). Let the radius of the sphere be R. Determine the work done as a particle of mass m is moved from the "north pole" of the sphere to a distance h above it ($h > R$). (The principle of superposition tells us the field of two mass distributions is just the vector sum of the field due to each.)

Problem 9.28 The gravitational field in some region of space is given by $\mathbf{g} = -kr^3\hat{\mathbf{r}}$, where k is a constant. What is the mass density ρ? What is ρ if the field is given by $\mathbf{g} = -(k/r^2)\hat{\mathbf{r}}$?

Problem 9.29 Consider a point P in the plane of a ring of mass M and radius a. Assume P lies a distance $r > a$ from the center of the ring. Determine the potential at r to an accuracy of order a^3/r^3. (Hint: To evaluate the integral use the binomial expansion and discard terms of order a^3/r^3 and higher.)

Problem 9.30 A collision between Earth and an asteroid of diameter 200 m would cause widespread damage and loss of life. Space scientists are interested in devising ways to deflect an approaching asteroid. Not too long ago, two NASA engineers[13] suggested placing a spacecraft near such an asteroid and using the gravitational attraction between spacecraft and asteroid to tow it away. This "gravitational tractor" would hover a distance d from the asteroid. Its engines would be directed as shown in Figure 9.13 so that they would not blast the surface of the asteroid. The angle ϕ is the half angle of the rocket jet; you may assume a value of 20° for this angle. (a) Show that the minimum thrust required to just balance the gravitational attraction of the asteroid is

$$T = \frac{GMm/d^2}{\cos[\sin^{-1}(r/d) + \phi]},$$

where r is the radius of the asteroid. (b) Evaluate T for a 200 m diameter asteroid ($r = 100$ m), having a density of 2×10^3 kg/m^3, if a 20 tonne spacecraft hovers at $d = 2r$. (c) Show that the velocity change imparted to the asteroid per second is $\Delta v = Gm/d^2$.

[13] E. T. Lu and S. G. Love, "Gravitational Tractor for Towing Asteroids," *Nature*, **438**, 177–178 (2005).

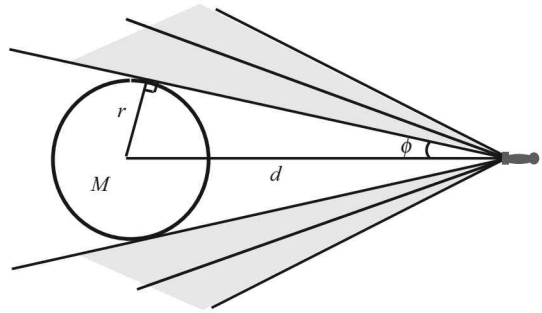


Figure 9.13 A "gravitational tractor." The shaded areas are the exhaust plumes from the rocket motors which are tilted away from the asteroid so as not to push on it. The angle ϕ is the half angle of the plume.

Problem 9.31 Show that the Minkowski metric yields $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$.

Computational Projects

Computational Technique: Numerical Integration

Two simple numerical integration schemes are called the "trapezoidal rule" and "Simpson's rule." These are two applications of a more general method called the "Newton–Cotes technique." Both schemes assume you know the value of the integrand at the end points and at a number of intermediate points.

The trapezoidal rule assumes the integrand $f(x)$ varies linearly between the "data" points. Thus, if we know $f(x)$ at x_i and at x_{i+1}, then

$$\int_{x_i}^{x_{i+1}} f(x)dx = \frac{1}{2}(x_{i+1} - x_i)(f_{i+1} + f_i).$$

The total integral from initial to final values of x will be the sum of such terms. (Note that the area under the $f(x)$ curve is broken up into trapezoids. The results are often not very accurate unless the x values are very closely spaced.)

Simpson's rule uses a quadratic fit to a series of three equally spaced points, separated by distances d. Then

$$\int_{x_i}^{x_{i+1}} f(x)dx = \frac{d}{3}(f_i + 4f_{i+1} + f_{i+2}).$$

Note that the number of values of x must be odd.

Computational Project 9.1 A sphere of radius 1 m has a density that varies as $\rho(r) = 5r^3$ for $0 \leq r \leq 1$. Obtain the mass of the sphere analytically and numerically with the trapezoidal rule and Simpson's rule for nine equally spaced values of r. Compare results of the three calculations.

Computational Project 9.2 Write a computer program to determine the position and velocity of a projectile fired vertically from the Earth's surface with an initial speed of 3000 m/s. Plot the position and velocity as a function of time. The Earth is assumed to be a sphere. You may neglect air resistance, the rotation of the Earth, and the effect of any other astronomical bodies, but note that the acceleration due to gravity varies with distance from the center of the Earth.

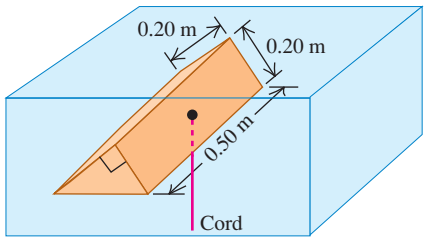
Computational Project 9.3 Plot the potential for two equal point masses. (To make things easier, assume you are using a system of units in which the masses are unity, they are separated by unit distance and the gravitational constant is unity.) You can let the two particles lie on the x-axis.

Computational Project 9.4 Imagine a planet in the shape of a cube with side 10^5 m and density 1000 kg/m^3. Draw a Cartesian coordinate system with origin at the center of the cube and axes passing through the centers of the faces. Obtain an integral expression for the potential at a point 10^6 m from the origin along one of the axes. Write a computer program to carry out the integration and obtain a numerical answer.

upward. (c) Derive an expression for the tension in the cord when the elevator is accelerating *downward* with an acceleration of magnitude a. Calculate the tension when $a = 2.50\ \text{m/s}^2$ downward. (d) What is the tension when the elevator is in free fall with a downward acceleration equal to g?

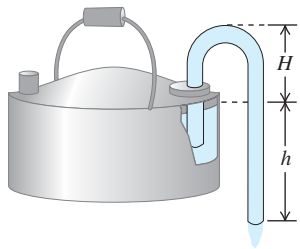
12.97 ••• **CALC** Suppose a piece of styrofoam, $\rho = 180\ \text{kg/m}^3$, is held completely submerged in water (Fig. P12.97). (a) What is the tension in the cord? Find this using Archimedes's principle. (b) Use $p = p_0 + \rho g h$ to calculate directly the force exerted by the water on the two sloped sides and the bottom of the styrofoam; then show that the vector sum of these forces is the buoyant force.

Figure **P12.97**



12.98 ••• A *siphon,* as shown in Fig. P12.98, is a convenient device for removing liquids from containers. To establish the flow, the tube must be initially filled with fluid. Let the fluid have density ρ, and let the atmospheric pressure be p_{atm}. Assume that the cross-sectional area of the tube is the same at all points along it. (a) If the lower end of the siphon is at a distance h below the surface of the liquid in the container, what is the speed of the fluid as it flows out the lower end of the siphon? (Assume that the container has a very large diameter, and ignore any effects of viscosity.) (b) A curious feature of a siphon is that the fluid initially flows "uphill." What is the greatest height H that the high point of the tube can have if flow is still to occur?

Figure **P12.98**



Answers

Chapter Opening Question ?

The flesh of both the shark and the tropical fish is denser than seawater, so left to themselves they would sink. However, a tropical fish has a gas-filled body cavity called a swimbladder, so that the *average* density of the fish's body is the same as that of seawater and the fish neither sinks nor rises. Sharks have no such cavity. Hence they must swim constantly to keep from sinking, using their pectoral fins to provide lift much like the wings of an airplane (see Section 12.5).

Test Your Understanding Questions

12.1 Answer: (ii), (iv), (i) and (iii) (tie), (v) In each case the average density equals the mass divided by the volume. Hence we have (i) $\rho = (4.00\ \text{kg})/(1.60 \times 10^{-3}\ \text{m}^3) = 2.50 \times 10^3\ \text{kg/m}^3$; (ii) $\rho = (8.00\ \text{kg})/(1.60 \times 10^{-3}\ \text{m}^3) = 5.00 \times 10^3\ \text{kg/m}^3$; (iii) $\rho = (8.00\ \text{kg})/(3.20 \times 10^{-3}\ \text{m}^3) = 2.50 \times 10^3\ \text{kg/m}^3$; (iv) $\rho = (2560\ \text{kg})/(0.640\ \text{m}^3) = 4.00 \times 10^3\ \text{kg/m}^3$; (v) $\rho = (2560\ \text{kg})/(1.28\ \text{m}^3) = 2.00 \times 10^3\ \text{kg/m}^3$. Note that compared to object (i), object (ii) has double the mass but the same volume and so has double the average density. Object (iii) has double the mass and double the volume of object (i), so (i) and (iii) have the same average density. Finally, object (v) has the same mass as object (iv) but double the volume, so (v) has half the average density of (iv).

12.2 Answer: (ii) From Eq. (12.9), the pressure outside the barometer is equal to the product $\rho g h$. When the barometer is taken out of the refrigerator, the density ρ decreases while the height h of the mercury column remains the same. Hence the air pressure must be lower outdoors than inside the refrigerator.

12.3 Answer: (i) Consider the water, the statue, and the container together as a system; the total weight of the system does not depend on whether the statue is immersed. The total supporting force, including the tension T and the upward force F of the scale on the container (equal to the scale reading), is the same in both cases. But we saw in Example 12.5 that T decreases by 7.84 N when the statue is immersed, so the scale reading F must *increase* by 7.84 N. An alternative viewpoint is that the water exerts an upward buoyant force of 7.84 N on the statue, so the statue must exert an equal downward force on the water, making the scale reading 7.84 N greater than the weight of water and container.

12.4 Answer: (ii) A highway that narrows from three lanes to one is like a pipe whose cross-sectional area narrows to one-third of its value. If cars behaved like the molecules of an incompressible fluid, then as the cars encountered the one-lane section, the spacing between cars (the "density") would stay the same but the cars would triple their speed. This would keep the "volume flow rate" (number of cars per second passing a point on the highway) the same. In real life cars behave like the molecules of a *compressible* fluid: They end up packed closer (the "density" increases) and fewer cars per second pass a point on the highway (the "volume flow rate" decreases).

12.5 Answer: (ii) Newton's second law tells us that a body accelerates (its velocity changes) in response to a net force. In fluid flow, a pressure difference between two points means that fluid particles moving between those two points experience a force, and this force causes the fluid particles to accelerate and change speed.

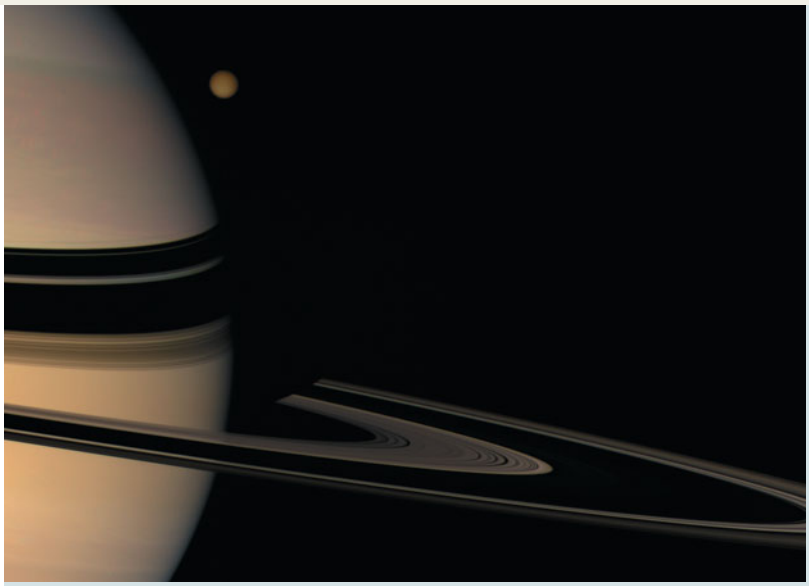
12.6 Answer: (iv) The required pressure is proportional to $1/R^4$, where R is the inside radius of the needle (half the inside diameter). With the smaller-diameter needle, the pressure is greater by a factor of $[(0.60\ \text{mm})/(0.30\ \text{mm})]^4 = 2^4 = 16$.

Bridging Problem

Answers: **(a)** $y = H - \left(\dfrac{d}{D}\right)^2 \sqrt{2gH}\, t + \left(\dfrac{d}{D}\right)^4 \dfrac{gt^2}{2}$

(b) $T = \sqrt{\dfrac{2H}{g}}\left(\dfrac{D}{d}\right)^2$ **(c)** $\sqrt{2}$

13 GRAVITATION



? The rings of Saturn are made of countless individual orbiting particles. Do all the ring particles orbit at the same speed, or do the inner particles orbit faster or slower than the outer ones?

LEARNING GOALS

By studying this chapter, you will learn:

- How to calculate the gravitational forces that any two bodies exert on each other.
- How to relate the weight of an object to the general expression for gravitational force.
- How to use and interpret the generalized expression for gravitational potential energy.
- How to relate the speed, orbital period, and mechanical energy of a satellite in a circular orbit.
- The laws that describe the motions of planets, and how to work with these laws.
- What black holes are, how to calculate their properties, and how they are discovered.

Some of the earliest investigations in physical science started with questions that people asked about the night sky. Why doesn't the moon fall to earth? Why do the planets move across the sky? Why doesn't the earth fly off into space rather than remaining in orbit around the sun? The study of gravitation provides the answers to these and many related questions.

As we remarked in Chapter 5, gravitation is one of the four classes of interactions found in nature, and it was the earliest of the four to be studied extensively. Newton discovered in the 17th century that the same interaction that makes an apple fall out of a tree also keeps the planets in their orbits around the sun. This was the beginning of *celestial mechanics,* the study of the dynamics of objects in space. Today, our knowledge of celestial mechanics allows us to determine how to put a satellite into any desired orbit around the earth or to choose just the right trajectory to send a spacecraft to another planet.

In this chapter you will learn the basic law that governs gravitational interactions. This law is *universal:* Gravity acts in the same fundamental way between the earth and your body, between the sun and a planet, and between a planet and one of its moons. We'll apply the law of gravitation to phenomena such as the variation of weight with altitude, the orbits of satellites around the earth, and the orbits of planets around the sun.

13.1 Newton's Law of Gravitation

The example of gravitational attraction that's probably most familiar to you is your *weight,* the force that attracts you toward the earth. During his study of the motions of the planets and of the moon, Newton discovered the fundamental character of the gravitational attraction between *any* two bodies. Along with his

three laws of motion, Newton published the **law of gravitation** in 1687. It may be stated as follows:

Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

Translating this into an equation, we have

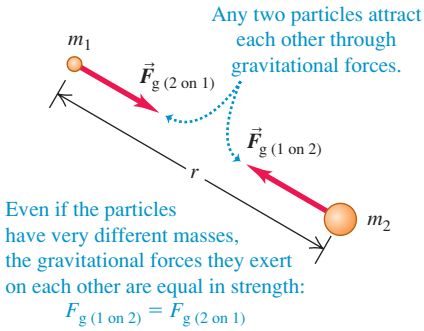
$$F_g = \frac{Gm_1m_2}{r^2} \qquad \text{(law of gravitation)} \qquad (13.1)$$

where F_g is the magnitude of the gravitational force on either particle, m_1 and m_2 are their masses, r is the distance between them (Fig. 13.1), and G is a fundamental physical constant called the **gravitational constant.** The numerical value of G depends on the system of units used.

Equation (13.1) tells us that the gravitational force between two particles decreases with increasing distance r: If the distance is doubled, the force is only one-fourth as great, and so on. Although many of the stars in the night sky are far more massive than the sun, they are so far away that their gravitational force on the earth is negligibly small.

CAUTION **Don't confuse g and G** Because the symbols g and G are so similar, it's common to confuse the two very different gravitational quantities that these symbols represent. Lowercase g is the acceleration due to gravity, which relates the weight w of a body to its mass m: $w = mg$. The value of g is different at different locations on the earth's surface and on the surfaces of different planets. By contrast, capital G relates the gravitational force between any two bodies to their masses and the distance between them. We call G a *universal* constant because it has the same value for any two bodies, no matter where in space they are located. In the next section we'll see how the values of g and G are related.

13.1 The gravitational forces between two particles of masses m_1 and m_2.



Gravitational forces always act along the line joining the two particles, and they form an action–reaction pair. Even when the masses of the particles are different, the two interaction forces have equal magnitude (Fig. 13.1). The attractive force that your body exerts on the earth has the same magnitude as the force that the earth exerts on you. When you fall from a diving board into a swimming pool, the entire earth rises up to meet you! (You don't notice this because the earth's mass is greater than yours by a factor of about 10^{23}. Hence the earth's acceleration is only 10^{-23} as great as yours.)

Gravitation and Spherically Symmetric Bodies

We have stated the law of gravitation in terms of the interaction between two *particles.* It turns out that the gravitational interaction of any two bodies having *spherically symmetric* mass distributions (such as solid spheres or spherical shells) is the same as though we concentrated all the mass of each at its center, as in Fig. 13.2. Thus, if we model the earth as a spherically symmetric body with mass m_E, the force it exerts on a particle or a spherically symmetric body with mass m, at a distance r between centers, is

$$F_g = \frac{Gm_Em}{r^2} \qquad (13.2)$$

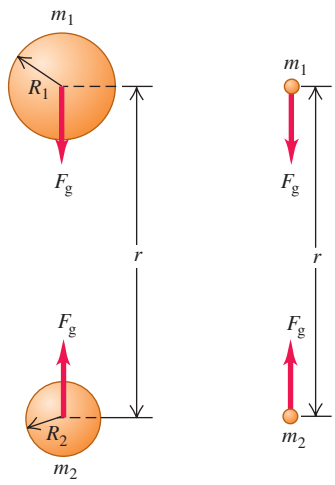
provided that the body lies outside the earth. A force of the same magnitude is exerted *on* the earth by the body. (We will prove these statements in Section 13.6.)

At points *inside* the earth the situation is different. If we could drill a hole to the center of the earth and measure the gravitational force on a body at various depths, we would find that toward the center of the earth the force *decreases,*

13.2 The gravitational effect *outside* any spherically symmetric mass distribution is the same as though all of the mass were concentrated at its center.

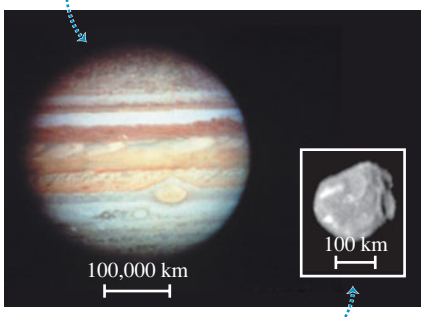
(a) The gravitational force between two spherically symmetric masses m_1 and m_2 ...

(b) ... is the same as if we concentrated all the mass of each sphere at the sphere's center.



13.3 Spherical and nonspherical bodies: the planet Jupiter and one of Jupiter's small moons, Amalthea.

Jupiter's mass is very large (1.90×10^{27} kg), so the mutual gravitational attraction of its parts has pulled it into a nearly spherical shape.



Amalthea, one of Jupiter's small moons, has a relatively tiny mass (7.17×10^{18} kg, only about 3.8×10^{-9} the mass of Jupiter) and weak mutual gravitation, so it has an irregular shape.

rather than increasing as $1/r^2$. As the body enters the interior of the earth (or other spherical body), some of the earth's mass is on the side of the body opposite from the center and pulls in the opposite direction. Exactly at the center, the earth's gravitational force on the body is zero.

Spherically symmetric bodies are an important case because moons, planets, and stars all tend to be spherical. Since all particles in a body gravitationally attract each other, the particles tend to move to minimize the distance between them. As a result, the body naturally tends to assume a spherical shape, just as a lump of clay forms into a sphere if you squeeze it with equal forces on all sides. This effect is greatly reduced in celestial bodies of low mass, since the gravitational attraction is less, and these bodies tend *not* to be spherical (Fig. 13.3).

Determining the Value of G

To determine the value of the gravitational constant G, we have to *measure* the gravitational force between two bodies of known masses m_1 and m_2 at a known distance r. The force is extremely small for bodies that are small enough to be brought into the laboratory, but it can be measured with an instrument called a *torsion balance,* which Sir Henry Cavendish used in 1798 to determine G.

Figure 13.4 shows a modern version of the Cavendish torsion balance. A light, rigid rod shaped like an inverted T is supported by a very thin, vertical quartz fiber. Two small spheres, each of mass m_1, are mounted at the ends of the horizontal arms of the T. When we bring two large spheres, each of mass m_2, to the positions shown, the attractive gravitational forces twist the T through a small angle. To measure this angle, we shine a beam of light on a mirror fastened to the T. The reflected beam strikes a scale, and as the T twists, the reflected beam moves along the scale.

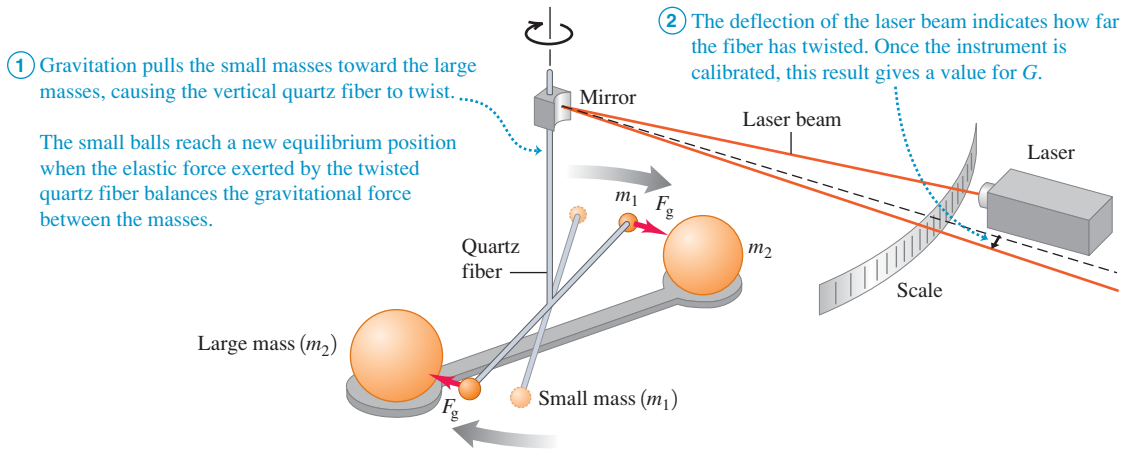
After calibrating the Cavendish balance, we can measure gravitational forces and thus determine G. The presently accepted value is

$$G = 6.67428(67) \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$$

To three significant figures, $G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$. Because $1\ \mathrm{N} = 1\ \mathrm{kg \cdot m/s^2}$, the units of G can also be expressed as $\mathrm{m^3/(kg \cdot s^2)}$.

Gravitational forces combine vectorially. If each of two masses exerts a force on a third, the *total* force on the third mass is the vector sum of the individual forces of the first two. Example 13.3 makes use of this property, which is often called *superposition of forces.*

13.4 The principle of the Cavendish balance, used for determining the value of G. The angle of deflection has been exaggerated here for clarity.



Example 13.1 Calculating gravitational force

The mass m_1 of one of the small spheres of a Cavendish balance is 0.0100 kg, the mass m_2 of the nearest large sphere is 0.500 kg, and the center-to-center distance between them is 0.0500 m. Find the gravitational force F_g on each sphere due to the other.

SOLUTION

IDENTIFY, SET UP, and EXECUTE: Because the spheres are spherically symmetric, we can calculate F_g by treating them as *particles* separated by 0.0500 m, as in Fig. 13.2. Each sphere experiences the same magnitude of force from the other sphere. We use Newton's law of gravitation, Eq. (13.1), to determine F_g:

$$F_g = \frac{(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(0.0100\ \text{kg})(0.500\ \text{kg})}{(0.0500\ \text{m})^2}$$
$$= 1.33 \times 10^{-10}\ \text{N}$$

EVALUATE: It's remarkable that such a small force could be measured—or even detected—more than 200 years ago. Only a very massive object such as the earth exerts a gravitational force we can feel.

Example 13.2 Acceleration due to gravitational attraction

Suppose the two spheres in Example 13.1 are placed with their centers 0.0500 m apart at a point in space far removed from all other bodies. What is the magnitude of the acceleration of each, relative to an inertial system?

SOLUTION

IDENTIFY, SET UP, and EXECUTE: Each sphere exerts on the other a gravitational force of the same magnitude F_g, which we found in Example 13.1. We can neglect any other forces. The *acceleration* magnitudes a_1 and a_2 are different because the masses are different. To determine these we'll use Newton's second law:

$$a_1 = \frac{F_g}{m_1} = \frac{1.33 \times 10^{-10}\ \text{N}}{0.0100\ \text{kg}} = 1.33 \times 10^{-8}\ \text{m/s}^2$$
$$a_2 = \frac{F_g}{m_2} = \frac{1.33 \times 10^{-10}\ \text{N}}{0.500\ \text{kg}} = 2.66 \times 10^{-10}\ \text{m/s}^2$$

EVALUATE: The larger sphere has 50 times the mass of the smaller one and hence has $\frac{1}{50}$ the acceleration. These accelerations are *not* constant; the gravitational forces increase as the spheres move toward each other.

Example 13.3 Superposition of gravitational forces

Many stars belong to *systems* of two or more stars held together by their mutual gravitational attraction. Figure 13.5 shows a three-star system at an instant when the stars are at the vertices of a 45° right triangle. Find the total gravitational force exerted on the small star by the two large ones.

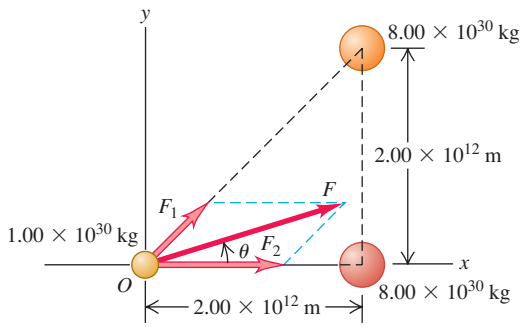
SOLUTION

IDENTIFY, SET UP, and EXECUTE: We use the principle of superposition: The total force $\vec{F}$ on the small star is the vector sum of the forces $\vec{F}_1$ and $\vec{F}_2$ due to each large star, as Fig. 13.5 shows. We assume that the stars are spheres as in Fig. 13.2. We first calculate the magnitudes F_1 and F_2 using Eq. (13.1) and then compute the vector sum using components:

$$F_1 = \frac{\left[\begin{array}{c}(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2) \\ \times\ (8.00 \times 10^{30}\ \text{kg})(1.00 \times 10^{30}\ \text{kg})\end{array}\right]}{(2.00 \times 10^{12}\ \text{m})^2 + (2.00 \times 10^{12}\ \text{m})^2}$$
$$= 6.67 \times 10^{25}\ \text{N}$$

$$F_2 = \frac{\left[\begin{array}{c}(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2) \\ \times\ (8.00 \times 10^{30}\ \text{kg})(1.00 \times 10^{30}\ \text{kg})\end{array}\right]}{(2.00 \times 10^{12}\ \text{m})^2}$$
$$= 1.33 \times 10^{26}\ \text{N}$$

13.5 The total gravitational force on the small star (at O) is the vector sum of the forces exerted on it by the two larger stars. (For comparison, the mass of the sun—a rather ordinary star—is 1.99×10^{30} kg and the earth–sun distance is 1.50×10^{11} m.)



The x- and y-components of these forces are

$$F_{1x} = (6.67 \times 10^{25}\ \text{N})(\cos 45°) = 4.72 \times 10^{25}\ \text{N}$$
$$F_{1y} = (6.67 \times 10^{25}\ \text{N})(\sin 45°) = 4.72 \times 10^{25}\ \text{N}$$
$$F_{2x} = 1.33 \times 10^{26}\ \text{N}$$
$$F_{2y} = 0$$

Continued

The components of the total force $\vec{F}$ on the small star are

$$F_x = F_{1x} + F_{2x} = 1.81 \times 10^{26}\,\text{N}$$
$$F_y = F_{1y} + F_{2y} = 4.72 \times 10^{25}\,\text{N}$$

The magnitude of $\vec{F}$ and its angle θ (see Fig. 13.5) are

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(1.81 \times 10^{26}\,\text{N})^2 + (4.72 \times 10^{25}\,\text{N})^2}$$
$$= 1.87 \times 10^{26}\,\text{N}$$
$$\theta = \arctan\frac{F_y}{F_x} = \arctan\frac{4.72 \times 10^{25}\,\text{N}}{1.81 \times 10^{26}\,\text{N}} = 14.6°$$

EVALUATE: While the force magnitude F is tremendous, the magnitude of the resulting acceleration is not: $a = F/m = (1.87 \times 10^{26}\,\text{N})/(1.00 \times 10^{30}\,\text{kg}) = 1.87 \times 10^{-4}\,\text{m/s}^2$. Furthermore, the force $\vec{F}$ is *not* directed toward the center of mass of the two large stars.

13.6 Our solar system is part of a spiral galaxy like this one, which contains roughly 10^{11} stars as well as gas, dust, and other matter. The entire assemblage is held together by the mutual gravitational attraction of all the matter in the galaxy.



Why Gravitational Forces Are Important

Comparing Examples 13.1 and 13.3 shows that gravitational forces are negligible between ordinary household-sized objects, but very substantial between objects that are the size of stars. Indeed, gravitation is *the* most important force on the scale of planets, stars, and galaxies (Fig. 13.6). It is responsible for holding our earth together and for keeping the planets in orbit about the sun. The mutual gravitational attraction between different parts of the sun compresses material at the sun's core to very high densities and temperatures, making it possible for nuclear reactions to take place there. These reactions generate the sun's energy output, which makes it possible for life to exist on earth and for you to read these words.

The gravitational force is so important on the cosmic scale because it acts *at a distance,* without any direct contact between bodies. Electric and magnetic forces have this same remarkable property, but they are less important on astronomical scales because large accumulations of matter are electrically neutral; that is, they contain equal amounts of positive and negative charge. As a result, the electric and magnetic forces between stars or planets are very small or zero. The strong and weak interactions that we discussed in Section 5.5 also act at a distance, but their influence is negligible at distances much greater than the diameter of an atomic nucleus (about 10^{-14} m).

A useful way to describe forces that act at a distance is in terms of a *field.* One body sets up a disturbance or field at all points in space, and the force that acts on a second body at a particular point is its response to the first body's field at that point. There is a field associated with each force that acts at a distance, and so we refer to gravitational fields, electric fields, magnetic fields, and so on. We won't need the field concept for our study of gravitation in this chapter, so we won't discuss it further here. But in later chapters we'll find that the field concept is an extraordinarily powerful tool for describing electric and magnetic interactions.

Test Your Understanding of Section 13.1 The planet Saturn has about 100 times the mass of the earth and is about 10 times farther from the sun than the earth is. Compared to the acceleration of the earth caused by the sun's gravitational pull, how great is the acceleration of Saturn due to the sun's gravitation? (i) 100 times greater; (ii) 10 times greater; (iii) the same; (iv) $\frac{1}{10}$ as great; (v) $\frac{1}{100}$ as great.

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13.2 Weight

We defined the *weight* of a body in Section 4.4 as the attractive gravitational force exerted on it by the earth. We can now broaden our definition:

The weight of a body is the total gravitational force exerted on the body by all other bodies in the universe.

When the body is near the surface of the earth, we can neglect all other gravitational forces and consider the weight as just the earth's gravitational attraction. At the surface of the *moon* we consider a body's weight to be the gravitational attraction of the moon, and so on.

If we again model the earth as a spherically symmetric body with radius R_E and mass m_E, the weight w of a small body of mass m at the earth's surface (a distance R_E from its center) is

$$w = F_g = \frac{Gm_E m}{R_E^2} \qquad \text{(weight of a body of mass } m \text{ at the earth's surface)} \tag{13.3}$$

But we also know from Section 4.4 that the weight w of a body is the force that causes the acceleration g of free fall, so by Newton's second law, $w = mg$. Equating this with Eq. (13.3) and dividing by m, we find

$$g = \frac{Gm_E}{R_E^2} \qquad \text{(acceleration due to gravity at the earth's surface)} \tag{13.4}$$

The acceleration due to gravity g is independent of the mass m of the body because m doesn't appear in this equation. We already knew that, but we can now see how it follows from the law of gravitation.

We can *measure* all the quantities in Eq. (13.4) except for m_E, so this relationship allows us to compute the mass of the earth. Solving Eq. (13.4) for m_E and using $R_E = 6380\text{ km} = 6.38 \times 10^6\text{ m}$ and $g = 9.80\text{ m/s}^2$, we find

$$m_E = \frac{gR_E^2}{G} = 5.98 \times 10^{24}\text{ kg}$$

This is very close to the currently accepted value of 5.974×10^{24} kg. Once Cavendish had measured G, he computed the mass of the earth in just this way.

At a point above the earth's surface a distance r from the center of the earth (a distance $r - R_E$ above the surface), the weight of a body is given by Eq. (13.3) with R_E replaced by r:

$$w = F_g = \frac{Gm_E m}{r^2} \tag{13.5}$$

The weight of a body decreases inversely with the square of its distance from the earth's center (Fig. 13.7). Figure 13.8 shows how the weight varies with height above the earth for an astronaut who weighs 700 N at the earth's surface.

The *apparent* weight of a body on earth differs slightly from the earth's gravitational force because the earth rotates and is therefore not precisely an inertial frame of reference. We have ignored this effect in our earlier discussion and have assumed that the earth *is* an inertial system. We will return to the effect of the earth's rotation in Section 13.7.

While the earth is an approximately spherically symmetric distribution of mass, it is *not* uniform throughout its volume. To demonstrate this, let's first calculate the average *density,* or mass per unit volume, of the earth. If we assume a spherical earth, the volume is

$$V_E = \tfrac{4}{3}\pi R_E^3 = \tfrac{4}{3}\pi(6.38 \times 10^6\text{ m})^3 = 1.09 \times 10^{21}\text{ m}^3$$

Application Walking and Running on the Moon

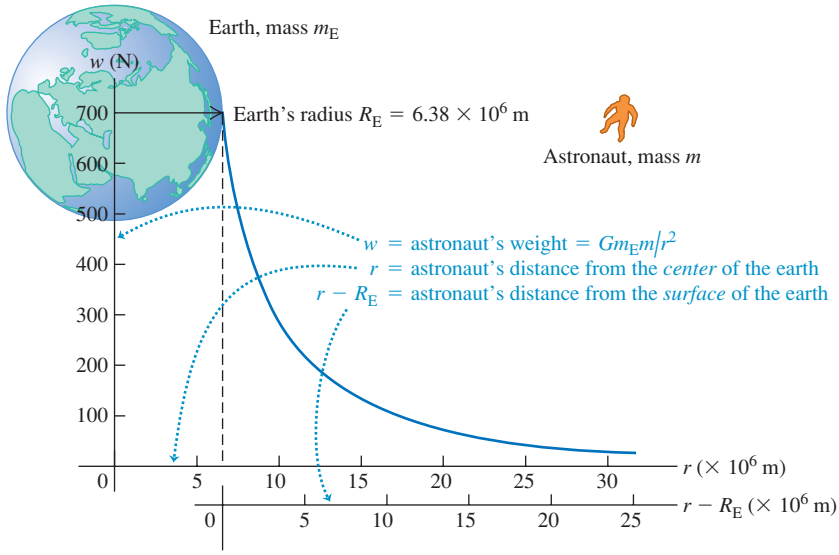
You automatically transition from a walk to a run when the vertical force you exert on the ground—which, by Newton's third law, equals the vertical force the ground exerts on you—exceeds your weight. This transition from walking to running happens at much lower speeds on the moon, where objects weigh only 17% as much as on earth. Hence, the Apollo astronauts found themselves running even when moving relatively slowly during their moon "walks."



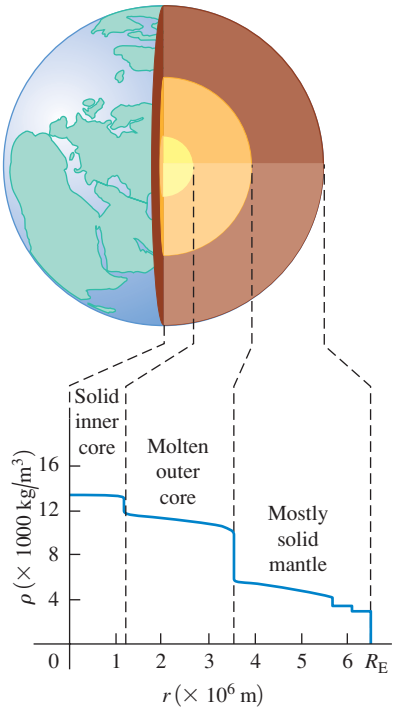
13.7 In an airliner at high altitude, you are farther from the center of the earth than when on the ground and hence weigh slightly less. Can you show that at an altitude of 10 km above the surface, you weigh 0.3% less than you do on the ground?



13.8 An astronaut who weighs 700 N at the earth's surface experiences less gravitational attraction when above the surface. The relevant distance r is from the astronaut to the *center* of the earth (*not* from the astronaut to the earth's surface).



13.9 The density of the earth decreases with increasing distance from its center.



The average density ρ (the Greek letter rho) of the earth is the total mass divided by the total volume:

$$\rho = \frac{m_E}{V_E} = \frac{5.97 \times 10^{24}\ \text{kg}}{1.09 \times 10^{21}\ \text{m}^3}$$
$$= 5500\ \text{kg/m}^3 = 5.5\ \text{g/cm}^3$$

(For comparison, the density of water is 1000 kg/m^3 = 1.00 g/cm^3.) If the earth were uniform, we would expect rocks near the earth's surface to have this same density. In fact, the density of surface rocks is substantially lower, ranging from about 2000 kg/m^3 for sedimentary rocks to about 3300 kg/m^3 for basalt. So the earth *cannot* be uniform, and the interior of the earth must be much more dense than the surface in order that the *average* density be 5500 kg/m^3. According to geophysical models of the earth's interior, the maximum density at the center is about 13,000 kg/m^3. Figure 13.9 is a graph of density as a function of distance from the center.

Example 13.4 Gravity on Mars

A robotic lander with an earth weight of 3430 N is sent to Mars, which has radius $R_M = 3.40 \times 10^6$ m and mass $m_M = 6.42 \times 10^{23}$ kg (see Appendix F). Find the weight F_g of the lander on the Martian surface and the acceleration there due to gravity, g_M.

SOLUTION

IDENTIFY and SET UP: To find F_g we use Eq. (13.3), replacing m_E and R_E with m_M and R_M. We determine the lander mass m from the lander's earth weight w and then find g_M from $F_g = mg_M$.

EXECUTE: The lander's earth weight is $w = mg$, so

$$m = \frac{w}{g} = \frac{3430\ \text{N}}{9.80\ \text{m/s}^2} = 350\ \text{kg}$$

The mass is the same no matter where the lander is. From Eq. (13.3), the lander's weight on Mars is

$$F_g = \frac{Gm_Mm}{R_M^2}$$
$$= \frac{(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(6.42 \times 10^{23}\ \text{kg})(350\ \text{kg})}{(3.40 \times 10^6\ \text{m})^2}$$
$$= 1.30 \times 10^3\ \text{N}$$

The acceleration due to gravity on Mars is

$$g_M = \frac{F_g}{m} = \frac{1.30 \times 10^3\ \text{N}}{350\ \text{kg}} = 3.7\ \text{m/s}^2$$

EVALUATE: Even though Mars has just 11% of the earth's mass (6.42×10^{23} kg versus 5.98×10^{24} kg), the acceleration due to gravity g_M (and hence an object's weight F_g) is roughly 40% as large as on earth. That's because g_M is also inversely proportional to the square of the planet's radius, and Mars has only 53% the radius of earth (3.40×10^6 m versus 6.38×10^6 m).

You can check our result for g_M by using Eq. (13.4), with appropriate replacements. Do you get the same answer?

Test Your Understanding of Section 13.2 Rank the following hypothetical planets in order from highest to lowest value of g at the surface:
(i) mass = 2 times the mass of the earth, radius = 2 times the radius of the earth;
(ii) mass = 4 times the mass of the earth, radius = 4 times the radius of the earth;
(iii) mass = 4 times the mass of the earth, radius = 2 times the radius of the earth;
(iv) mass = 2 times the mass of the earth, radius = 4 times the radius of the earth. I

13.3 Gravitational Potential Energy

When we first introduced gravitational potential energy in Section 7.1, we assumed that the gravitational force on a body is constant in magnitude and direction. This led to the expression $U = mgy$. But the earth's gravitational force on a body of mass m at any point outside the earth is given more generally by Eq. (13.2), $F_g = Gm_E m/r^2$, where m_E is the mass of the earth and r is the distance of the body from the earth's center. For problems in which r changes enough that the gravitational force can't be considered constant, we need a more general expression for gravitational potential energy.

To find this expression, we follow the same steps as in Section 7.1. We consider a body of mass m outside the earth, and first compute the work W_{grav} done by the gravitational force when the body moves directly away from or toward the center of the earth from $r = r_1$ to $r = r_2$, as in Fig. 13.10. This work is given by

$$W_{grav} = \int_{r_1}^{r_2} F_r\, dr \qquad (13.6)$$

where F_r is the radial component of the gravitational force $\vec{F}$—that is, the component in the direction *outward* from the center of the earth. Because $\vec{F}$ points directly *inward* toward the center of the earth, F_r is negative. It differs from Eq. (13.2), the magnitude of the gravitational force, by a minus sign:

$$F_r = -\frac{Gm_E m}{r^2} \qquad (13.7)$$

Substituting Eq. (13.7) into Eq. (13.6), we see that W_{grav} is given by

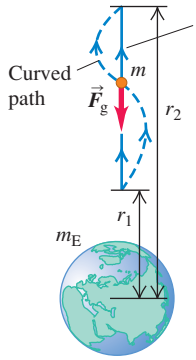
$$W_{grav} = -Gm_E m \int_{r_1}^{r_2} \frac{dr}{r^2} = \frac{Gm_E m}{r_2} - \frac{Gm_E m}{r_1} \qquad (13.8)$$

The path doesn't have to be a straight line; it could also be a curve like the one in Fig. 13.10. By an argument similar to that in Section 7.1, this work depends only on the initial and final values of r, not on the path taken. This also proves that the gravitational force is always *conservative*.

We now define the corresponding potential energy U so that $W_{grav} = U_1 - U_2$, as in Eq. (7.3). Comparing this with Eq. (13.8), we see that the appropriate definition for **gravitational potential energy** is

$$U = -\frac{Gm_E m}{r} \qquad \text{(gravitational potential energy)} \qquad (13.9)$$

13.10 Calculating the work done on a body by the gravitational force as the body moves from radial coordinate r_1 to r_2.



The gravitational force is conservative: The work done by $\vec{F}_g$ does not depend on the path taken from r_1 to r_2.

13.11 A graph of the gravitational potential energy U for the system of the earth (mass m_E) and an astronaut (mass m) versus the astronaut's distance r from the center of the earth.

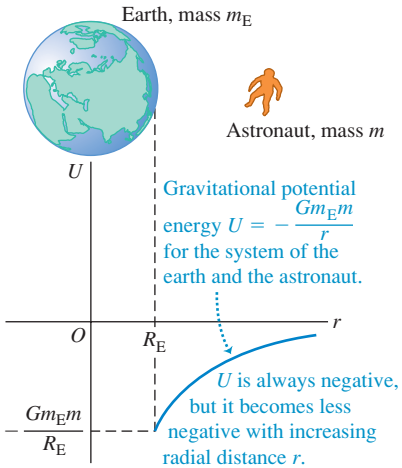


Figure 13.11 shows how the gravitational potential energy depends on the distance r between the body of mass m and the center of the earth. When the body moves away from the earth, r increases, the gravitational force does negative work, and U increases (i.e., becomes less negative). When the body "falls" toward earth, r decreases, the gravitational work is positive, and the potential energy decreases (i.e., becomes more negative).

You may be troubled by Eq. (13.9) because it states that gravitational potential energy is always negative. But in fact you've seen negative values of U before. In using the formula $U = mgy$ in Section 7.1, we found that U was negative whenever the body of mass m was at a value of y below the arbitrary height we chose to be $y = 0$—that is, whenever the body and the earth were closer together than some certain arbitrary distance. (See, for instance, Example 7.2 in Section 7.1.) In defining U by Eq. (13.9), we have chosen U to be zero when the body of mass m is infinitely far from the earth ($r = \infty$). As the body moves toward the earth, gravitational potential energy decreases and so becomes negative.

If we wanted, we could make $U = 0$ at the surface of the earth, where $r = R_E$, by simply adding the quantity Gm_Em/R_E to Eq. (13.9). This would make U positive when $r > R_E$. We won't do this for two reasons: One, it would make the expression for U more complicated; and two, the added term would not affect the *difference* in potential energy between any two points, which is the only physically significant quantity.

CAUTION **Gravitational force vs. gravitational potential energy** Be careful not to confuse the expressions for gravitational force, Eq. (13.7), and gravitational potential energy, Eq. (13.9). The force F_r is proportional to $1/r^2$, while potential energy U is proportional to $1/r$.

Armed with Eq. (13.9), we can now use general energy relationships for problems in which the $1/r^2$ behavior of the earth's gravitational force has to be included. If the gravitational force on the body is the only force that does work, the total mechanical energy of the system is constant, or *conserved.* In the following example we'll use this principle to calculate **escape speed,** the speed required for a body to escape completely from a planet.

Example 13.5 "From the earth to the moon"

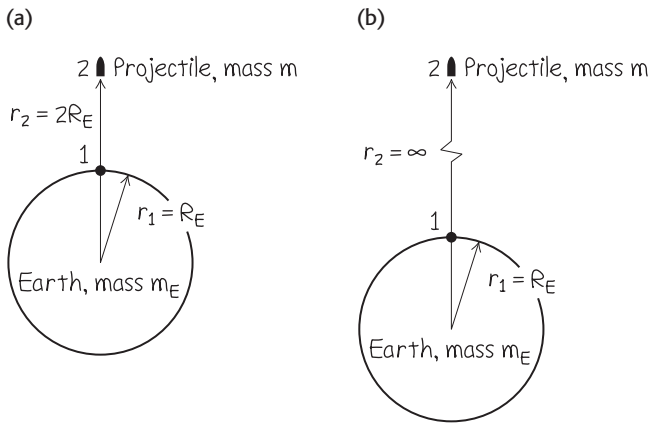
In Jules Verne's 1865 story with this title, three men went to the moon in a shell fired from a giant cannon sunk in the earth in Florida. (a) Find the minimum muzzle speed needed to shoot a shell straight up to a height above the earth equal to the earth's radius R_E. (b) Find the minimum muzzle speed that would allow a shell to escape from the earth completely (the *escape speed*). Neglect air resistance, the earth's rotation, and the gravitational pull of the moon. The earth's radius and mass are $R_E = 6.38 \times 10^6$ m and $m_E = 5.97 \times 10^{24}$ kg.

SOLUTION

IDENTIFY and SET UP: Once the shell leaves the cannon muzzle, only the (conservative) gravitational force does work. Hence we can use conservation of mechanical energy to find the speed at which the shell must leave the muzzle so as to come to a halt (a) at two earth radii from the earth's center and (b) at an infinite distance from earth. The energy-conservation equation is $K_1 + U_1 = K_2 + U_2$, with U given by Eq. (13.9).

Figure 13.12 shows our sketches. Point 1 is at $r_1 = R_E$, where the shell leaves the cannon with speed v_1 (the target variable). Point 2 is where the shell reaches its maximum height; in part (a) $r_2 = 2R_E$ (Fig. 13.12a), and in part (b) $r_2 = \infty$ (Fig 13.12b). In both cases $v_2 = 0$ and $K_2 = 0$. Let m be the mass of the shell (with passengers).

13.12 Our sketches for this problem.



EXECUTE: (a) We solve the energy-conservation equation for v_1:

$$K_1 + U_1 = K_2 + U_2$$

$$\tfrac{1}{2}mv_1^2 + \left(-\frac{Gm_Em}{R_E}\right) = 0 + \left(-\frac{Gm_Em}{2R_E}\right)$$

$$v_1 = \sqrt{\frac{Gm_E}{R_E}} = \sqrt{\frac{(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.97 \times 10^{24}\ \text{kg})}{6.38 \times 10^6\ \text{m}}}$$

$$= 7900\ \text{m/s}\ (= 28{,}400\ \text{km/h} = 17{,}700\ \text{mi/h})$$

(b) Now $r_2 = \infty$ so $U_2 = 0$ (see Fig. 13.11). Since $K_2 = 0$, the total mechanical energy $K_2 + U_2$ is zero in this case. Again we solve the energy-conservation equation for v_1:

$$\tfrac{1}{2}mv_1^2 + \left(-\frac{Gm_Em}{R_E}\right) = 0 + 0$$

$$v_1 = \sqrt{\frac{2Gm_E}{R_E}}$$

$$= \sqrt{\frac{2(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.97 \times 10^{24}\ \text{kg})}{6.38 \times 10^6\ \text{m}}}$$

$$= 1.12 \times 10^4\ \text{m/s}\ (= 40{,}200\ \text{km/h} = 25{,}000\ \text{mi/h})$$

EVALUATE: Our result in part (b) doesn't depend on the mass of the shell or the direction of launch. A modern spacecraft launched from Florida must attain essentially the speed found in part (b) to escape the earth; however, before launch it's already moving at 410 m/s to the east because of the earth's rotation. Launching to the east takes advantage of this "free" contribution toward escape speed.

To generalize, the initial speed v_1 needed for a body to escape from the surface of a spherical body of mass M and radius R (ignoring air resistance) is $v_1 = \sqrt{2GM/R}$ (escape speed). This equation yields escape speeds of 5.02×10^3 m/s for Mars, 5.95×10^4 m/s for Jupiter, and 6.18×10^5 m/s for the sun.

More on Gravitational Potential Energy

As a final note, let's show that when we are close to the earth's surface, Eq. (13.9) reduces to the familiar $U = mgy$ from Chapter 7. We first rewrite Eq. (13.8) as

$$W_{\text{grav}} = Gm_Em\frac{r_1 - r_2}{r_1 r_2}$$

If the body stays close to the earth, then in the denominator we may replace r_1 and r_2 by R_E, the earth's radius, so

$$W_{\text{grav}} = Gm_Em\frac{r_1 - r_2}{R_E^2}$$

According to Eq. (13.4), $g = Gm_E/R_E^2$, so

$$W_{\text{grav}} = mg(r_1 - r_2)$$

If we replace the r's by y's, this is just Eq. (7.1) for the work done by a constant gravitational force. In Section 7.1 we used this equation to derive Eq. (7.2), $U = mgy$, so we may consider Eq. (7.2) for gravitational potential energy to be a special case of the more general Eq. (13.9).

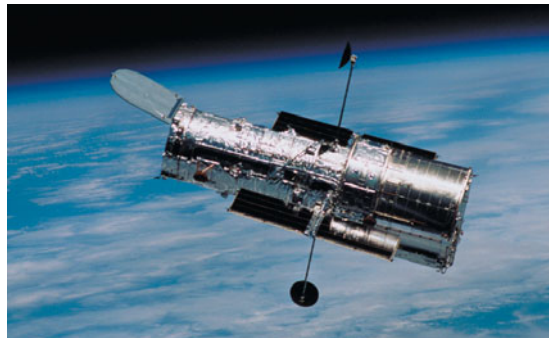
Test Your Understanding of Section 13.3 Is it possible for a planet to have the same surface gravity as the earth (that is, the same value of g at the surface) and yet have a greater escape speed?

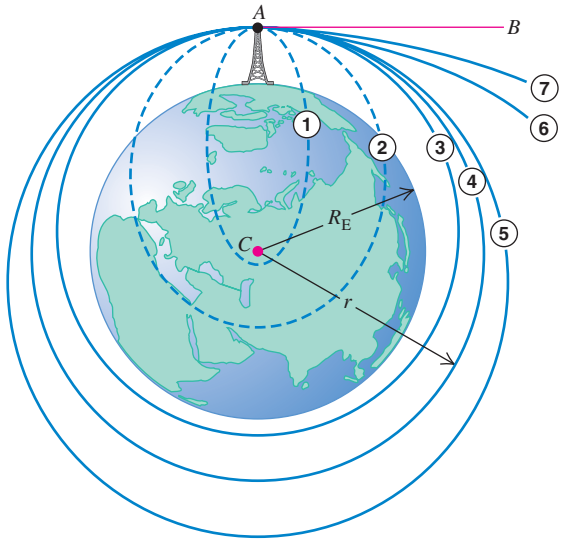
13.4 The Motion of Satellites

Artificial satellites orbiting the earth are a familiar part of modern technology (Fig. 13.13). But how do they stay in orbit, and what determines the properties of their orbits? We can use Newton's laws and the law of gravitation to provide the answers. We'll see in the next section that the motion of planets can be analyzed in the same way.

To begin, think back to the discussion of projectile motion in Section 3.3. In Example 3.6 a motorcycle rider rides horizontally off the edge of a cliff, launching himself into a parabolic path that ends on the flat ground at the base of the cliff. If he survives and repeats the experiment with increased launch speed, he will land farther from the starting point. We can imagine him launching himself with great enough speed that the earth's curvature becomes significant. As he falls, the earth curves away beneath him. If he is going fast enough, and if his

13.13 With a length of 13.2 m and a mass of 11,000 kg, the Hubble Space Telescope is among the largest satellites placed in orbit.





A projectile is launched from *A* toward *B*. Trajectories (1) through (7) show the effect of increasing initial speed.

13.14 Trajectories of a projectile launched from a great height (ignoring air resistance). Orbits 1 and 2 would be completed as shown if the earth were a point mass at *C*. (This illustration is based on one in Isaac Newton's *Principia*.)

launch point is high enough that he clears the mountaintops, he may be able to go right on around the earth without ever landing.

Figure 13.14 shows a variation on this theme. We launch a projectile from point *A* in the direction *AB*, tangent to the earth's surface. Trajectories 1 through 7 show the effect of increasing the initial speed. In trajectories 3 through 5 the projectile misses the earth and becomes a satellite. If there is no retarding force, the projectile's speed when it returns to point *A* is the same as its initial speed and it repeats its motion indefinitely.

Trajectories 1 through 5 close on themselves and are called **closed orbits.** All closed orbits are ellipses or segments of ellipses; trajectory 4 is a circle, a special case of an ellipse. (We'll discuss the properties of an ellipse in Section 13.5.) Trajectories 6 and 7 are **open orbits.** For these paths the projectile never returns to its starting point but travels ever farther away from the earth.

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Satellites: Circular Orbits

A *circular* orbit, like trajectory 4 in Fig. 13.14, is the simplest case. It is also an important case, since many artificial satellites have nearly circular orbits and the orbits of the planets around the sun are also fairly circular. The only force acting on a satellite in circular orbit around the earth is the earth's gravitational attraction, which is directed toward the center of the earth and hence toward the center of the orbit (Fig. 13.15). As we discussed in Section 5.4, this means that the satellite is in *uniform* circular motion and its speed is constant. The satellite isn't falling *toward* the earth; rather, it's constantly falling *around* the earth. In a circular orbit the speed is just right to keep the distance from the satellite to the center of the earth constant.

Let's see how to find the constant speed v of a satellite in a circular orbit. The radius of the orbit is r, measured from the *center* of the earth; the acceleration of the satellite has magnitude $a_{rad} = v^2/r$ and is always directed toward the center of the circle. By the law of gravitation, the net force (gravitational force) on the satellite of mass m has magnitude $F_g = Gm_E m/r^2$ and is in the same direction as the acceleration. Newton's second law ($\Sigma\vec{F} = m\vec{a}$) then tells us that

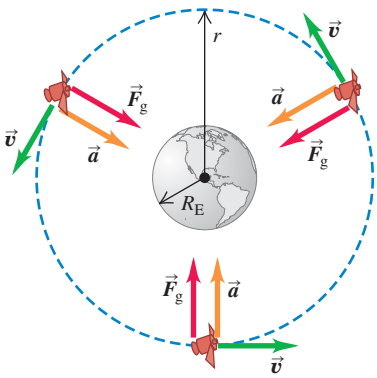
$$\frac{Gm_E m}{r^2} = \frac{mv^2}{r}$$

Solving this for v, we find

$$v = \sqrt{\frac{Gm_E}{r}} \qquad \text{(circular orbit)} \qquad (13.10)$$

This relationship shows that we can't choose the orbit radius r and the speed v independently; for a given radius r, the speed v for a circular orbit is determined.

13.15 The force $\vec{F}_g$ due to the earth's gravitational attraction provides the centripetal acceleration that keeps a satellite in orbit. Compare to Fig. 5.28.



The satellite is in a circular orbit: Its acceleration $\vec{a}$ is always perpendicular to its velocity $\vec{v}$, so its speed v is constant.

The satellite's mass m doesn't appear in Eq. (13.10), which shows that the motion of a satellite does not depend on its mass. If we could cut a satellite in half without changing its speed, each half would continue on with the original motion. An astronaut on board a space shuttle is herself a satellite of the earth, held by the earth's gravitational attraction in the same orbit as the shuttle. The astronaut has the same velocity and acceleration as the shuttle, so nothing is pushing her against the floor or walls of the shuttle. She is in a state of *apparent weightlessness,* as in a freely falling elevator; see the discussion following Example 5.9 in Section 5.2. (*True* weightlessness would occur only if the astronaut were infinitely far from any other masses, so that the gravitational force on her would be zero.) Indeed, every part of her body is apparently weightless; she feels nothing pushing her stomach against her intestines or her head against her shoulders (Fig. 13.16).

Apparent weightlessness is not just a feature of circular orbits; it occurs whenever gravity is the only force acting on a spacecraft. Hence it occurs for orbits of any shape, including open orbits such as trajectories 6 and 7 in Fig. 13.14.

We can derive a relationship between the radius r of a circular orbit and the period T, the time for one revolution. The speed v is the distance $2\pi r$ traveled in one revolution, divided by the period:

$$v = \frac{2\pi r}{T} \tag{13.11}$$

To get an expression for T, we solve Eq. (13.11) for T and substitute v from Eq. (13.10):

$$T = \frac{2\pi r}{v} = 2\pi r\sqrt{\frac{r}{Gm_E}} = \frac{2\pi r^{3/2}}{\sqrt{Gm_E}} \qquad \text{(circular orbit)} \tag{13.12}$$

Equations (13.10) and (13.12) show that larger orbits correspond to slower speeds and longer periods. As an example, the International Space Station orbits 6800 km from the center of the earth (400 km above the earth's surface) with an orbital speed of 7.7 km/s and an orbital period of 93 minutes. The moon orbits the earth in a much larger orbit of radius 384,000 km, and so has a much slower orbital speed (1.0 km/s) and a much longer orbital period (27.3 days).

It's interesting to compare Eq. (13.10) to the calculation of escape speed in Example 13.5. We see that the escape speed from a spherical body with radius R is $\sqrt{2}$ times greater than the speed of a satellite in a circular orbit at that radius. If our spacecraft is in circular orbit around *any* planet, we have to multiply our speed by a factor of $\sqrt{2}$ to escape to infinity, regardless of the planet's mass.

Since the speed v in a circular orbit is determined by Eq. (13.10) for a given orbit radius r, the total mechanical energy $E = K + U$ is determined as well. Using Eqs. (13.9) and (13.10), we have

$$E = K + U = \tfrac{1}{2}mv^2 + \left(-\frac{Gm_Em}{r}\right) = \tfrac{1}{2}m\left(\frac{Gm_E}{r}\right) - \frac{Gm_Em}{r}$$

$$= -\frac{Gm_Em}{2r} \qquad \text{(circular orbit)} \tag{13.13}$$

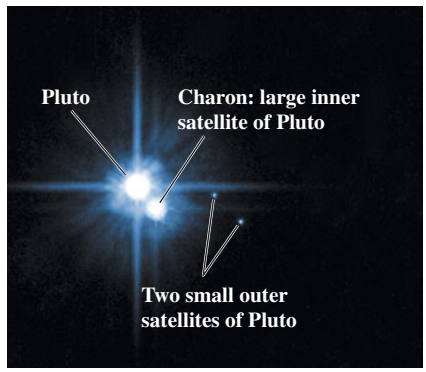
The total mechanical energy in a circular orbit is negative and equal to one-half the potential energy. Increasing the orbit radius r means increasing the mechanical energy (that is, making E less negative). If the satellite is in a relatively low orbit that encounters the outer fringes of earth's atmosphere, mechanical energy decreases due to negative work done by the force of air resistance; as a result, the orbit radius decreases until the satellite hits the ground or burns up in the atmosphere.

We have talked mostly about earth satellites, but we can apply the same analysis to the circular motion of *any* body under its gravitational attraction to a stationary body. Other examples include the earth's moon and the moons of other worlds (Fig. 13.17).

13.16 These space shuttle astronauts are in a state of apparent weightlessness. Which are right side up and which are upside down?



13.17 The two small satellites of the minor planet Pluto were discovered in 2005. In accordance with Eqs. (13.10) and (13.12), the satellite in the larger orbit has a slower orbital speed and a longer orbital period than the satellite in the smaller orbit.



Example 13.6 A satellite orbit

You wish to put a 1000-kg satellite into a circular orbit 300 km above the earth's surface. (a) What speed, period, and radial acceleration will it have? (b) How much work must be done to the satellite to put it in orbit? (c) How much additional work would have to be done to make the satellite escape the earth? The earth's radius and mass are given in Example 13.5 (Section 13.3).

SOLUTION

IDENTIFY and SET UP: The satellite is in a circular orbit, so we can use the equations derived in this section. In part (a), we first find the radius r of the satellite's orbit from its altitude. We then calculate the speed v and period T using Eqs. (13.10) and (13.12) and the acceleration from $a_{\text{rad}} = v^2/r$. In parts (b) and (c), the work required is the difference between the initial and final mechanical energy, which for a circular orbit is given by Eq. (13.13).

EXECUTE: (a) The radius of the satellite's orbit is $r = 6380\text{ km} + 300\text{ km} = 6680\text{ km} = 6.68 \times 10^6\text{ m}$. From Eq. (13.10), the orbital speed is

$$v = \sqrt{\frac{Gm_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.97 \times 10^{24}\text{ kg})}{6.68 \times 10^6\text{ m}}}$$
$$= 7720\text{ m/s}$$

We find the orbital period from Eq. (13.12):

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.68 \times 10^6\text{ m})}{7720\text{ m/s}} = 5440\text{ s} = 90.6\text{ min}$$

Finally, the radial acceleration is

$$a_{\text{rad}} = \frac{v^2}{r} = \frac{(7720\text{ m/s})^2}{6.68 \times 10^6\text{ m}} = 8.92\text{ m/s}^2$$

This is the value of g at a height of 300 km above the earth's surface; it is about 10% less than the value of g at the surface.

(b) The work required is the difference between E_2, the total mechanical energy when the satellite is in orbit, and E_1, the total mechanical energy when the satellite was at rest on the launch pad. From Eq. (13.13), the energy in orbit is

$$E_2 = -\frac{Gm_E m}{2r}$$
$$= -\frac{(6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.97 \times 10^{24}\text{ kg})(1000\text{ kg})}{2(6.68 \times 10^6\text{ m})}$$
$$= -2.98 \times 10^{10}\text{ J}$$

The satellite's kinetic energy is zero on the launch pad ($r = R_E$), so

$$E_1 = K_1 + U_1 = 0 + \left(-\frac{Gm_E m}{R_E}\right)$$
$$= -\frac{(6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.97 \times 10^{24}\text{ kg})(1000\text{ kg})}{6.38 \times 10^6\text{ m}}$$
$$= -6.24 \times 10^{10}\text{ J}$$

Hence the work required is

$$W_{\text{required}} = E_2 - E_1 = (-2.98 \times 10^{10}\text{ J}) - (-6.24 \times 10^{10}\text{ J})$$
$$= 3.26 \times 10^{10}\text{ J}$$

(c) We saw in part (b) of Example 13.5 that the minimum total mechanical energy for a satellite to escape to infinity is zero. Here, the total mechanical energy in the circular orbit is $E_2 = -2.98 \times 10^{10}$ J; to increase this to zero, an amount of work equal to 2.98×10^{10} J would have to be done on the satellite, presumably by rocket engines attached to it.

EVALUATE: In part (b) we ignored the satellite's initial kinetic energy (while it was still on the launch pad) due to the rotation of the earth. How much difference does this make? (See Example 13.5 for useful data.)

Test Your Understanding of Section 13.4 Your personal spacecraft is in a low-altitude circular orbit around the earth. Air resistance from the outer regions of the atmosphere does negative work on the spacecraft, causing the orbital radius to decrease slightly. Does the speed of the spacecraft (i) remain the same, (ii) increase, or (iii) decrease?

13.5 Kepler's Laws and the Motion of Planets

The name *planet* comes from a Greek word meaning "wanderer," and indeed the planets continuously change their positions in the sky relative to the background of stars. One of the great intellectual accomplishments of the 16th and 17th centuries was the threefold realization that the earth is also a planet, that all planets orbit the sun, and that the apparent motions of the planets as seen from the earth can be used to precisely determine their orbits.

The first and second of these ideas were published by Nicolaus Copernicus in Poland in 1543. The nature of planetary orbits was deduced between 1601 and 1619 by the German astronomer and mathematician Johannes Kepler, using a voluminous set of precise data on apparent planetary motions compiled by his mentor, the Danish astronomer Tycho Brahe. By trial and error, Kepler

discovered three empirical laws that accurately described the motions of the planets:

1. **Each planet moves in an elliptical orbit, with the sun at one focus of the ellipse.**
2. **A line from the sun to a given planet sweeps out equal areas in equal times.**
3. **The periods of the planets are proportional to the $\frac{3}{2}$ powers of the major axis lengths of their orbits.**

Kepler did not know *why* the planets moved in this way. Three generations later, when Newton turned his attention to the motion of the planets, he discovered that each of Kepler's laws can be *derived;* they are consequences of Newton's laws of motion and the law of gravitation. Let's see how each of Kepler's laws arises.

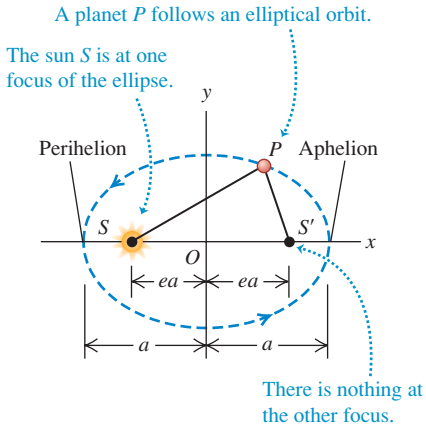
Kepler's First Law

First consider the elliptical orbits described in Kepler's first law. Figure 13.18 shows the geometry of an ellipse. The longest dimension is the *major axis,* with half-length a; this half-length is called the **semi-major axis.** The sum of the distances from S to P and from S' to P is the same for all points on the curve. S and S' are the *foci* (plural of *focus*). The sun is at S, and the planet is at P; we think of them both as points because the size of each is very small in comparison to the distance between them. There is nothing at the other focus S'.

The distance of each focus from the center of the ellipse is ea, where e is a dimensionless number between 0 and 1 called the **eccentricity.** If $e = 0$, the ellipse is a circle. The actual orbits of the planets are fairly circular; their eccentricities range from 0.007 for Venus to 0.206 for Mercury. (The earth's orbit has $e = 0.017$.) The point in the planet's orbit closest to the sun is the *perihelion,* and the point most distant from the sun is the *aphelion.*

Newton was able to show that for a body acted on by an attractive force proportional to $1/r^2$, the only possible closed orbits are a circle or an ellipse; he also showed that open orbits (trajectories 6 and 7 in Fig. 13.14) must be parabolas or hyperbolas. These results can be derived by a straightforward application of Newton's laws and the law of gravitation, together with a lot more differential equations than we're ready for.

13.18 Geometry of an ellipse. The sum of the distances SP and $S'P$ is the same for every point on the curve. The sizes of the sun (S) and planet (P) are exaggerated for clarity.



Kepler's Second Law

Figure 13.19 shows Kepler's second law. In a small time interval dt, the line from the sun S to the planet P turns through an angle $d\theta$. The area swept out is the colored triangle with height r, base length $r\,d\theta$, and area $dA = \frac{1}{2}r^2\,d\theta$ in Fig. 13.19b. The rate at which area is swept out, dA/dt, is called the *sector velocity:*

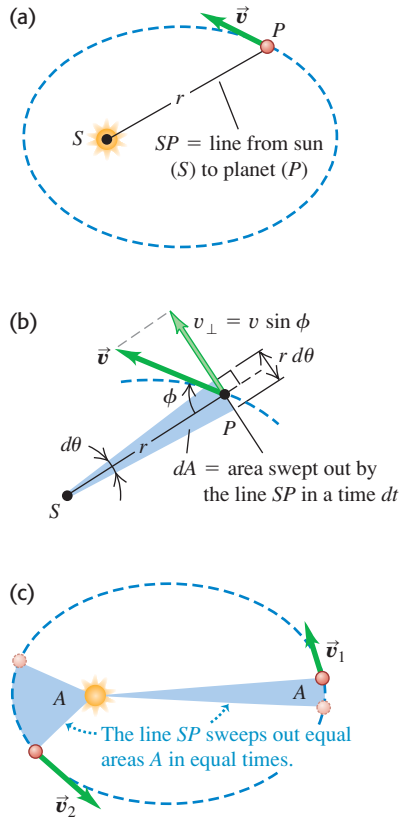
$$\frac{dA}{dt} = \tfrac{1}{2}r^2\frac{d\theta}{dt} \qquad (13.14)$$

The essence of Kepler's second law is that the sector velocity has the same value at all points in the orbit. When the planet is close to the sun, r is small and $d\theta/dt$ is large; when the planet is far from the sun, r is large and $d\theta/dt$ is small.

To see how Kepler's second law follows from Newton's laws, we express dA/dt in terms of the velocity vector $\vec{v}$ of the planet P. The component of $\vec{v}$ perpendicular to the radial line is $v_\perp = v\sin\phi$. From Fig. 13.19b the displacement along the direction of $v_\perp$ during time dt is $r\,d\theta$, so we also have $v_\perp = r\,d\theta/dt$. Using this relationship in Eq. (13.14), we find

$$\frac{dA}{dt} = \tfrac{1}{2}rv\sin\phi \qquad \text{(sector velocity)} \qquad (13.15)$$

13.19 (a) The planet (P) moves about the sun (S) in an elliptical orbit. (b) In a time dt the line SP sweeps out an area $dA = \frac{1}{2}(r\,d\theta)r = \frac{1}{2}r^2\,d\theta$. (c) The planet's speed varies so that the line SP sweeps out the same area A in a given time t regardless of the planet's position in its orbit.



Now $rv\sin\phi$ is the magnitude of the vector product $\vec{r} \times \vec{v}$, which in turn is $1/m$ times the angular momentum $\vec{L} = \vec{r} \times m\vec{v}$ of the planet with respect to the sun. So we have

$$\frac{dA}{dt} = \frac{1}{2m}|\vec{r} \times m\vec{v}| = \frac{L}{2m} \qquad (13.16)$$

Thus Kepler's second law—that sector velocity is constant—means that angular momentum is constant!

It is easy to see why the angular momentum of the planet *must* be constant. According to Eq. (10.26), the rate of change of $\vec{L}$ equals the torque of the gravitational force $\vec{F}$ acting on the planet:

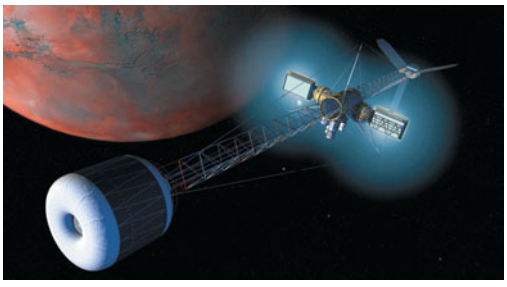
$$\frac{d\vec{L}}{dt} = \vec{\tau} = \vec{r} \times \vec{F}$$

In our situation, $\vec{r}$ is the vector from the sun to the planet, and the force $\vec{F}$ is directed from the planet to the sun. So these vectors always lie along the same line, and their vector product $\vec{r} \times \vec{F}$ is zero. Hence $d\vec{L}/dt = \mathbf{0}$. This conclusion does not depend on the $1/r^2$ behavior of the force; angular momentum is conserved for *any* force that acts always along the line joining the particle to a fixed point. Such a force is called a *central force*. (Kepler's first and third laws are valid *only* for a $1/r^2$ force.)

Conservation of angular momentum also explains why the orbit lies in a plane. The vector $\vec{L} = \vec{r} \times m\vec{v}$ is always perpendicular to the plane of the vectors $\vec{r}$ and $\vec{v}$; since $\vec{L}$ is constant in magnitude *and* direction, $\vec{r}$ and $\vec{v}$ always lie in the same plane, which is just the plane of the planet's orbit.

Application Biological Hazards of Interplanetary Travel

A spacecraft sent from earth to another planet spends most of its journey coasting along an elliptical orbit with the sun at one focus. Rockets are used only at the start and end of the journey, and even the trip to a nearby planet like Mars takes several months. During its journey, the spacecraft is exposed to cosmic rays—radiation that emanates from elsewhere in our galaxy. (On earth we're shielded from this radiation by our planet's magnetic field, as we'll describe in Chapter 27.) This poses no problem for a robotic spacecraft, but would be a severe medical hazard for astronauts undertaking such a voyage.



Kepler's Third Law

We have already derived Kepler's third law for the particular case of circular orbits. Equation (13.12) shows that the period of a satellite or planet in a circular orbit is proportional to the $\frac{3}{2}$ power of the orbit radius. Newton was able to show that this same relationship holds for an *elliptical* orbit, with the orbit radius r replaced by the semi-major axis a:

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_S}} \qquad \text{(elliptical orbit around the sun)} \qquad (13.17)$$

Since the planet orbits the sun, not the earth, we have replaced the earth's mass m_E in Eq. (13.12) with the sun's mass m_S. Note that the period does not depend on the eccentricity e. An asteroid in an elongated elliptical orbit with semi-major axis a will have the same orbital period as a planet in a circular orbit of radius a. The key difference is that the asteroid moves at different speeds at different points in its elliptical orbit (Fig. 13.19c), while the planet's speed is constant around its circular orbit.

Conceptual Example 13.7 Orbital speeds

At what point in an elliptical orbit (see Fig. 13.19) does a planet move the fastest? The slowest?

SOLUTION

Mechanical energy is conserved as a planet moves in its orbit. The planet's kinetic energy $K = \frac{1}{2}mv^2$ is maximum when the potential energy $U = -Gm_Sm/r$ is minimum (that is, most negative; see Fig. 13.11), which occurs when the sun–planet distance r is a minimum. Hence the speed v is greatest at perihelion. Similarly, K is minimum when r is maximum, so the speed is slowest at aphelion.

Your intuition about falling bodies is helpful here. As the planet falls inward toward the sun, it picks up speed, and its speed is maximum when closest to the sun. The planet slows down as it moves away from the sun, and its speed is minimum at aphelion.

Example 13.8 Kepler's third law

The asteroid Pallas has an orbital period of 4.62 years and an orbital eccentricity of 0.233. Find the semi-major axis of its orbit.

SOLUTION

IDENTIFY and SET UP: This example uses Kepler's third law, which relates the period T and the semi-major axis a for an orbiting object (such as an asteroid). We use Eq. (13.17) to determine a; from Appendix F we have $m_S = 1.99 \times 10^{30}$ kg, and a conversion factor from Appendix E gives $T = (4.62\text{ yr})(3.156 \times 10^7\text{ s/yr}) = 1.46 \times 10^8$ s. Note that we don't need the value of the eccentricity.

EXECUTE: From Eq. (13.17), $a^{3/2} = [(Gm_S)^{1/2}T]/2\pi$. To solve for a, we raise both sides of this expression to the $\frac{2}{3}$ power and then substitute the values of G, m_S, and T:

$$a = \left(\frac{Gm_ST^2}{4\pi^2}\right)^{1/3} = 4.15 \times 10^{11}\text{ m}$$

(Plug in the numbers yourself to check.)

EVALUATE: Our result is intermediate between the semi-major axes of Mars and Jupiter (see Appendix F). Most known asteroids orbit in an "asteroid belt" between the orbits of these two planets.

Example 13.9 Comet Halley

Comet Halley moves in an elongated elliptical orbit around the sun (Fig. 13.20). Its distances from the sun at perihelion and aphelion are 8.75×10^7 km and 5.26×10^9 km, respectively. Find the orbital semi-major axis, eccentricity, and period.

SOLUTION

IDENTIFY and SET UP: We are to find the semi-major axis a, eccentricity e, and orbital period T. We can use Fig. 13.18 to find a and e from the given perihelion and aphelion distances. Knowing a, we can find T from Kepler's third law, Eq. (13.17).

EXECUTE: From Fig. 13.18, the length $2a$ of the major axis equals the sum of the comet–sun distance at perihelion and the comet–sun distance at aphelion. Hence

$$a = \frac{(8.75 \times 10^7\text{ km}) + (5.26 \times 10^9\text{ km})}{2} = 2.67 \times 10^9\text{ km}$$

Figure 13.19 also shows that the comet–sun distance at perihelion is $a - ea = a(1 - e)$. This distance is 8.75×10^7 km, so

$$e = 1 - \frac{8.75 \times 10^7\text{ km}}{a} = 1 - \frac{8.75 \times 10^7\text{ km}}{2.67 \times 10^9\text{ km}} = 0.967$$

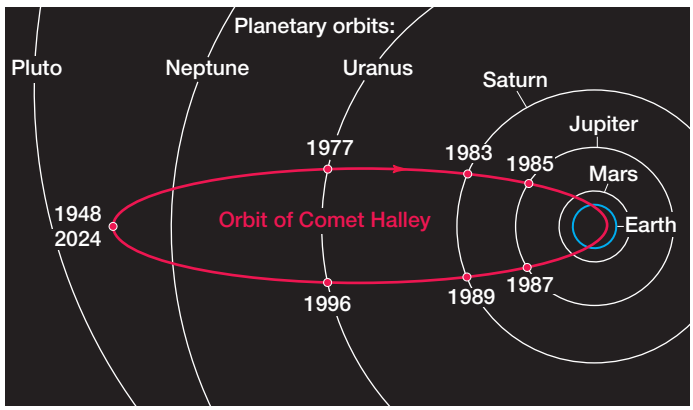
From Eq. (13.17), the period is

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_S}} = \frac{2\pi(2.67 \times 10^{12}\text{ m})^{3/2}}{\sqrt{(6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(1.99 \times 10^{30}\text{ kg})}}$$
$$= 2.38 \times 10^9\text{ s} = 75.5\text{ years}$$

EVALUATE: The eccentricity is close to 1, so the orbit is very elongated (see Fig. 13.20a). Comet Halley was at perihelion in early 1986 (Fig. 13.20b); it will next reach perihelion one period later, in 2061.

13.20 (a) The orbit of Comet Halley. (b) Comet Halley as it appeared in 1986. At the heart of the comet is an icy body, called the nucleus, that is about 10 km across. When the comet's orbit carries it close to the sun, the heat of sunlight causes the nucleus to partially evaporate. The evaporated material forms the tail, which can be tens of millions of kilometers long.

(a)



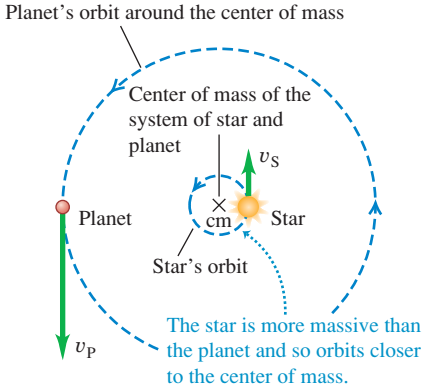
(b)



Planetary Motions and the Center of Mass

We have assumed that as a planet or comet orbits the sun, the sun remains absolutely stationary. Of course, this can't be correct; because the sun exerts a

13.21 A star and its planet both orbit about their common center of mass.

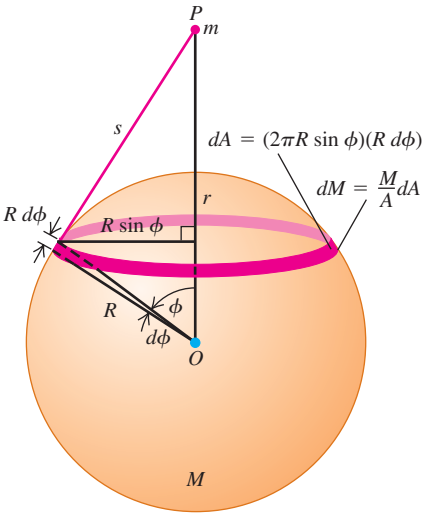


gravitational force on the planet, the planet exerts a gravitational force on the sun of the same magnitude but opposite direction. In fact, *both* the sun and the planet orbit around their common center of mass (Fig. 13.21). We've made only a small error by ignoring this effect, however; the sun's mass is about 750 times the total mass of all the planets combined, so the center of mass of the solar system is not far from the center of the sun. Remarkably, astronomers have used this effect to detect the presence of planets orbiting other stars. Sensitive telescopes are able to detect the apparent "wobble" of a star as it orbits the common center of mass of the star and an unseen companion planet. (The planets are too faint to observe directly.) By analyzing these "wobbles," astronomers have discovered planets in orbit around hundreds of other stars.

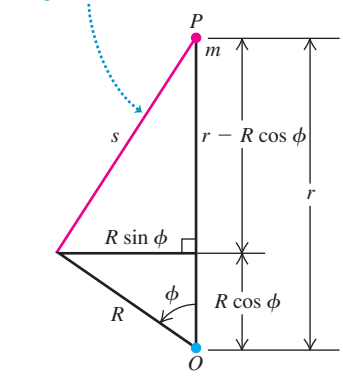
Newton's analysis of planetary motions is used on a daily basis by modern-day astronomers. But the most remarkable result of Newton's work is that the motions of bodies in the heavens obey the *same* laws of motion as do bodies on the earth. This *Newtonian synthesis,* as it has come to be called, is one of the great unifying principles of science. It has had profound effects on the way that humanity looks at the universe—not as a realm of impenetrable mystery, but as a direct extension of our everyday world, subject to scientific study and calculation.

13.22 Calculating the gravitational potential energy of interaction between a point mass m outside a spherical shell and a ring on the surface of the shell.

(a) Geometry of the situation



(b) The distance s is the hypotenuse of a right triangle with sides $(r - R\cos\phi)$ and $R\sin\phi$.



Test Your Understanding of Section 13.5 The orbit of Comet X has a semi-major axis that is four times longer than the semi-major axis of Comet Y. What is the ratio of the orbital period of X to the orbital period of Y? (i) 2; (ii) 4; (iii) 8; (iv) 16; (v) 32; (vi) 64. I

13.6 Spherical Mass Distributions

We have stated without proof that the gravitational interaction between two spherically symmetric mass distributions is the same as though all the mass of each were concentrated at its center. Now we're ready to prove this statement. Newton searched for a proof for several years, and he delayed publication of the law of gravitation until he found one.

Here's our program. Rather than starting with two spherically symmetric masses, we'll tackle the simpler problem of a point mass m interacting with a thin spherical shell with total mass M. We will show that when m is outside the sphere, the *potential energy* associated with this gravitational interaction is the same as though M were all concentrated at the center of the sphere. We learned in Section 7.4 that the force is the negative derivative of the potential energy, so the *force* on m is also the same as for a point mass M. Any spherically symmetric mass distribution can be thought of as being made up of many concentric spherical shells, so our result will also hold for *any* spherically symmetric M.

A Point Mass Outside a Spherical Shell

We start by considering a ring on the surface of the shell (Fig. 13.22a), centered on the line from the center of the shell to m. We do this because all of the particles that make up the ring are the same distance s from the point mass m. From Eq. (13.9) the potential energy of interaction between the earth (mass m_E) and a point mass m, separated by a distance r, is $U = -Gm_Em/r$. By changing notation in this expression, we see that the potential energy of interaction between the point mass m and a particle of mass m_i within the ring is given by

$$U_i = -\frac{Gmm_i}{s}$$

To find the potential energy of interaction between m and the entire ring of mass $dM = \sum_i m_i$, we sum this expression for U_i over all particles in the ring. Calling this potential energy dU, we find

$$dU = \sum_i U_i = \sum_i \left(-\frac{Gmm_i}{s} \right) = -\frac{Gm}{s} \sum_i m_i = -\frac{Gm\, dM}{s} \qquad (13.18)$$

To proceed, we need to know the mass dM of the ring. We can find this with the aid of a little geometry. The radius of the shell is R, so in terms of the angle ϕ shown in the figure, the radius of the ring is $R\sin\phi$, and its circumference is $2\pi R\sin\phi$. The width of the ring is $R\, d\phi$, and its area dA is approximately equal to its width times its circumference:

$$dA = 2\pi R^2 \sin\phi\, d\phi$$

The ratio of the ring mass dM to the total mass M of the shell is equal to the ratio of the area dA of the ring to the total area $A = 4\pi R^2$ of the shell:

$$\frac{dM}{M} = \frac{2\pi R^2 \sin\phi\, d\phi}{4\pi R^2} = \tfrac{1}{2}\sin\phi\, d\phi \qquad (13.19)$$

Now we solve Eq. (13.19) for dM and substitute the result into Eq. (13.18) to find the potential energy of interaction between the point mass m and the ring:

$$dU = -\frac{GMm\sin\phi\, d\phi}{2s} \qquad (13.20)$$

The total potential energy of interaction between the point mass and the *shell* is the integral of Eq. (13.20) over the whole sphere as ϕ varies from 0 to π (*not* 2π!) and s varies from $r - R$ to $r + R$. To carry out the integration, we have to express the integrand in terms of a single variable; we choose s. To express ϕ and $d\phi$ in terms of s, we have to do a little more geometry. Figure 13.22b shows that s is the hypotenuse of a right triangle with sides $(r - R\cos\phi)$ and $R\sin\phi$, so the Pythagorean theorem gives

$$\begin{aligned} s^2 &= (r - R\cos\phi)^2 + (R\sin\phi)^2 \\ &= r^2 - 2rR\cos\phi + R^2 \end{aligned} \qquad (13.21)$$

We take differentials of both sides:

$$2s\, ds = 2rR\sin\phi\, d\phi$$

Next we divide this by $2rR$ and substitute the result into Eq. (13.20):

$$dU = -\frac{GMm}{2s}\frac{s\, ds}{rR} = -\frac{GMm}{2rR}\, ds \qquad (13.22)$$

We can now integrate Eq. (13.22), recalling that s varies from $r - R$ to $r + R$:

$$U = -\frac{GMm}{2rR}\int_{r-R}^{r+R} ds = -\frac{GMm}{2rR}[(r + R) - (r - R)] \qquad (13.23)$$

Finally, we have

$$U = -\frac{GMm}{r} \qquad \text{(point mass } m \text{ outside spherical shell } M\text{)} \qquad (13.24)$$

This is equal to the potential energy of two point masses m and M at a distance r. So we have proved that the gravitational potential energy of the spherical shell M and the point mass m at any distance r is the same as though they were point masses. Because the force is given by $F_r = -dU/dr$, the force is also the same.

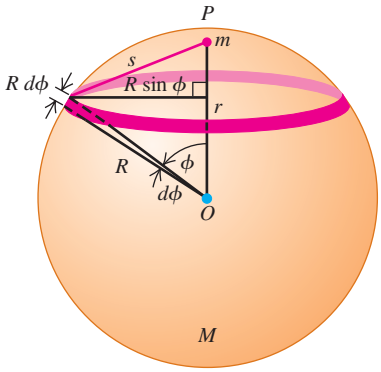
The Gravitational Force Between Spherical Mass Distributions

Any spherically symmetric mass distribution can be thought of as a combination of concentric spherical shells. Because of the principle of superposition of forces, what is true of one shell is also true of the combination. So we have proved half of what we set out to prove: that the gravitational interaction between any spherically symmetric mass distribution and a point mass is the same as though all the mass of the spherically symmetric distribution were concentrated at its center.

The other half is to prove that *two* spherically symmetric mass distributions interact as though they were both points. That's easier. In Fig. 13.22a the forces the two bodies exert on each other are an action–reaction pair, and they obey Newton's third law. So we have also proved that the force that m exerts *on* the sphere M is the same as though M were a point. But now if we replace m with a spherically symmetric mass distribution centered at m's location, the resulting gravitational force on any part of M is the same as before, and so is the total force. This completes our proof.

A Point Mass Inside a Spherical Shell

13.23 When a point mass m is *inside* a uniform spherical shell of mass M, the potential energy is the same no matter where inside the shell the point mass is located. The force from the masses' mutual gravitational interaction is zero.



We assumed at the beginning that the point mass m was outside the spherical shell, so our proof is valid only when m is outside a spherically symmetric mass distribution. When m is *inside* a spherical shell, the geometry is as shown in Fig. 13.23. The entire analysis goes just as before; Eqs. (13.18) through (13.22) are still valid. But when we get to Eq. (13.23), the limits of integration have to be changed to $R - r$ and $R + r$. We then have

$$U = -\frac{GMm}{2rR}\int_{R-r}^{R+r} ds = -\frac{GMm}{2rR}[(R + r) - (R - r)] \qquad (13.25)$$

and the final result is

$$U = -\frac{GMm}{R} \qquad \text{(point mass } m \text{ inside spherical shell } M\text{)} \qquad (13.26)$$

Compare this result to Eq. (13.24): Instead of having r, the distance between m and the center of M, in the denominator, we have R, the radius of the shell. This means that U in Eq. (13.26) doesn't depend on r and thus has the same value everywhere inside the shell. When m moves around inside the shell, no work is done on it, so the force on m at any point inside the shell must be zero.

More generally, at any point in the interior of any spherically symmetric mass distribution (not necessarily a shell), at a distance r from its center, the gravitational force on a point mass m is the same as though we removed all the mass at points farther than r from the center and concentrated all the remaining mass at the center.

Example 13.10 "Journey to the center of the earth"

Imagine that we drill a hole through the earth along a diameter and drop a mail pouch down the hole. Derive an expression for the gravitational force F_g on the pouch as a function of its distance from the earth's center. Assume that the earth's density is uniform (not a very realistic model; see Fig. 13.9).

SOLUTION

IDENTIFY and SET UP: From the discussion immediately above, the value of F_g at a distance r from the earth's center is determined only by the mass M within a spherical region of radius r (Fig. 13.24). Hence F_g is the same as if all the mass within radius r were concentrated at the center of the earth. The mass of a uniform sphere is proportional to the volume of the sphere, which is $\frac{4}{3}\pi r^3$ for a sphere of arbitrary radius r and $\frac{4}{3}\pi R_E^3$ for the entire earth.

EXECUTE: The ratio of the mass M of the sphere of radius r to the mass m_E of the earth is

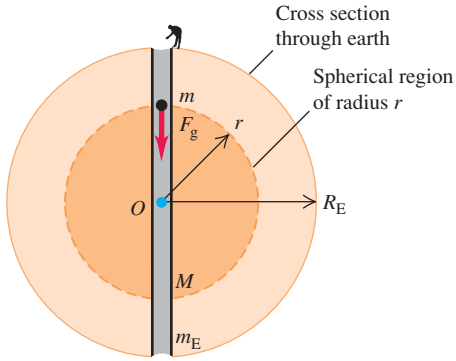
$$\frac{M}{m_E} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R_E^3} = \frac{r^3}{R_E^3} \qquad \text{so} \qquad M = m_E\frac{r^3}{R_E^3}$$

The magnitude of the gravitational force on m is then

$$F_g = \frac{GMm}{r^2} = \frac{Gm}{r^2}\left(m_E\frac{r^3}{R_E^3}\right) = \frac{Gm_E m}{R_E^3}r$$

EVALUATE: Inside this uniform-density sphere, F_g is *directly proportional* to the distance r from the center, rather than to $1/r^2$ as it is outside the sphere. At the surface $r = R_E$, we have $F_g = Gm_E m/R_E^2$, as we should. In the next chapter we'll learn how to compute the time it would take for the mail pouch to emerge on the other side of the earth.

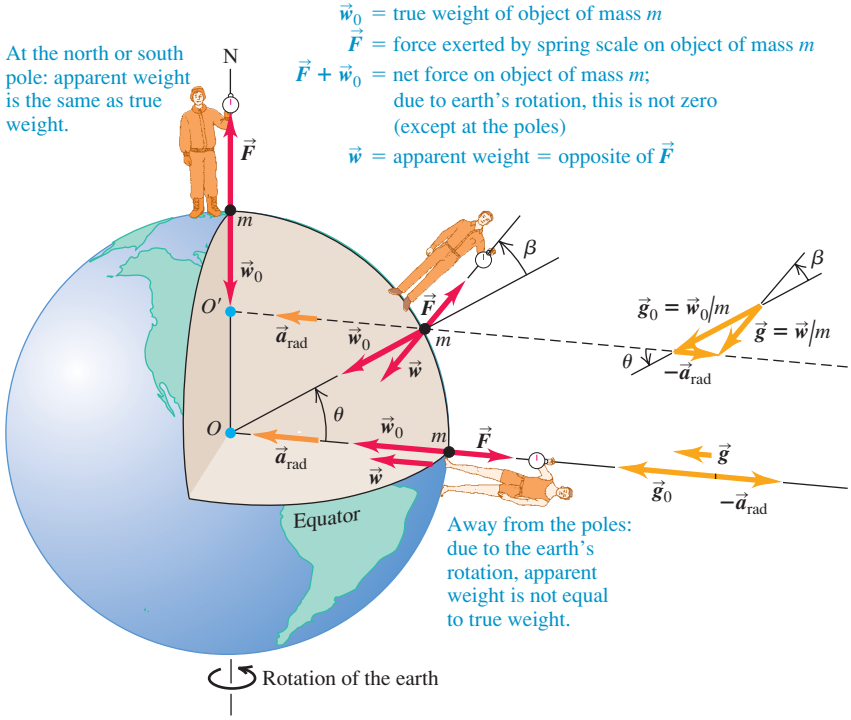
13.24 A hole through the center of the earth (assumed to be uniform). When an object is a distance r from the center, only the mass inside a sphere of radius r exerts a net gravitational force on it.



Test Your Understanding of Section 13.6 In the classic 1913 science-fiction novel *At the Earth's Core* by Edgar Rice Burroughs, explorers discover that the earth is a hollow sphere and that an entire civilization lives on the inside of the sphere. Would it be possible to stand and walk on the inner surface of a hollow, nonrotating planet?

13.7 Apparent Weight and the Earth's Rotation

Because the earth rotates on its axis, it is not precisely an inertial frame of reference. For this reason the apparent weight of a body on earth is not precisely equal to the earth's gravitational attraction, which we will call the **true weight** $\vec{w}_0$ of the body. Figure 13.25 is a cutaway view of the earth, showing three observers. Each one holds a spring scale with a body of mass m hanging from it. Each scale applies a tension force $\vec{F}$ to the body hanging from it, and the reading on each scale is the magnitude F of this force. If the observers are unaware of the earth's



13.25 Except at the poles, the reading for an object being weighed on a scale (the *apparent weight*) is less than the gravitational force of attraction on the object (the *true weight*). The reason is that a net force is needed to provide a centripetal acceleration as the object rotates with the earth. For clarity, the illustration greatly exaggerates the angle β between the true and apparent weight vectors.

rotation, each one *thinks* that the scale reading equals the weight of the body because he thinks the body on his spring scale is in equilibrium. So each observer thinks that the tension $\vec{F}$ must be opposed by an equal and opposite force $\vec{w}$, which we call the **apparent weight.** But if the bodies are rotating with the earth, they are *not* precisely in equilibrium. Our problem is to find the relationship between the apparent weight $\vec{w}$ and the true weight $\vec{w}_0$.

If we assume that the earth is spherically symmetric, then the true weight $\vec{w}_0$ has magnitude $Gm_E m/R_E^2$, where m_E and R_E are the mass and radius of the earth. This value is the same for all points on the earth's surface. If the center of the earth can be taken as the origin of an inertial coordinate system, then the body at the north pole really *is* in equilibrium in an inertial system, and the reading on that observer's spring scale is equal to w_0. But the body at the equator is moving in a circle of radius R_E with speed v, and there must be a net inward force equal to the mass times the centripetal acceleration:

$$w_0 - F = \frac{mv^2}{R_E}$$

So the magnitude of the apparent weight (equal to the magnitude of F) is

$$w = w_0 - \frac{mv^2}{R_E} \qquad \text{(at the equator)} \tag{13.27}$$

If the earth were not rotating, the body when released would have a free-fall acceleration $g_0 = w_0/m$. Since the earth *is* rotating, the falling body's actual acceleration relative to the observer at the equator is $g = w/m$. Dividing Eq. (13.27) by m and using these relationships, we find

$$g = g_0 - \frac{v^2}{R_E} \qquad \text{(at the equator)}$$

To evaluate v^2/R_E, we note that in 86,164 s a point on the equator moves a distance equal to the earth's circumference, $2\pi R_E = 2\pi(6.38 \times 10^6\ \text{m})$. (The solar day, 86,400 s, is $\frac{1}{365}$ longer than this because in one day the earth also completes $\frac{1}{365}$ of its orbit around the sun.) Thus we find

$$v = \frac{2\pi(6.38 \times 10^6\ \text{m})}{86{,}164\ \text{s}} = 465\ \text{m/s}$$

$$\frac{v^2}{R_E} = \frac{(465\ \text{m/s})^2}{6.38 \times 10^6\ \text{m}} = 0.0339\ \text{m/s}^2$$

So for a spherically symmetric earth the acceleration due to gravity should be about 0.03 m/s^2 less at the equator than at the poles.

At locations intermediate between the equator and the poles, the true weight $\vec{w}_0$ and the centripetal acceleration are not along the same line, and we need to write a vector equation corresponding to Eq. (13.27). From Fig. 13.25 we see that the appropriate equation is

$$\vec{w} = \vec{w}_0 - m\vec{a}_{\text{rad}} = m\vec{g}_0 - m\vec{a}_{\text{rad}} \tag{13.28}$$

The difference in the magnitudes of g and g_0 lies between zero and 0.0339 m/s^2. As shown in Fig. 13.25, the *direction* of the apparent weight differs from the direction toward the center of the earth by a small angle β, which is 0.1° or less.

Table 13.1 gives the values of g at several locations, showing variations with latitude. There are also small additional variations due to the lack of perfect spherical symmetry of the earth, local variations in density, and differences in elevation.

Table 13.1 Variations of g with Latitude and Elevation

Station	North Latitude	Elevation (m)	g (m/s^2)
Canal Zone	09°	0	9.78243
Jamaica	18°	0	9.78591
Bermuda	32°	0	9.79806
Denver, CO	40°	1638	9.79609
Pittsburgh, PA	40.5°	235	9.80118
Cambridge, MA	42°	0	9.80398
Greenland	70°	0	9.82534

Test Your Understanding of Section 13.7 Imagine a planet that has the same mass and radius as the earth, but that makes 10 rotations during the time the earth makes one rotation. What would be the difference between the acceleration due to gravity at the planet's equator and the acceleration due to gravity at its poles? (i) 0.00339 m/s^2; (ii) 0.0339 m/s^2; (iii) 0.339 m/s^2; (iv) 3.39 m/s^2.

13.8 Black Holes

The concept of a black hole is one of the most interesting and startling products of modern gravitational theory, yet the basic idea can be understood on the basis of Newtonian principles.

The Escape Speed from a Star

Think first about the properties of our own sun. Its mass $M = 1.99 \times 10^{30}$ kg and radius $R = 6.96 \times 10^8$ m are much larger than those of any planet, but compared to other stars, our sun is not exceptionally massive. You can find the sun's average density ρ in the same way we found the average density of the earth in Section 13.2:

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3} = \frac{1.99 \times 10^{30}\ \text{kg}}{\frac{4}{3}\pi(6.96 \times 10^8\ \text{m})^3} = 1410\ \text{kg/m}^3$$

The sun's temperatures range from 5800 K (about 5500°C or 10,000°F) at the surface up to 1.5×10^7 K (about 2.7×10^7°F) in the interior, so it surely contains no solids or liquids. Yet gravitational attraction pulls the sun's gas atoms together until the sun is, on average, 41% denser than water and about 1200 times as dense as the air we breathe.

Now think about the escape speed for a body at the surface of the sun. In Example 13.5 (Section 13.3) we found that the escape speed from the surface of a spherical mass M with radius R is $v = \sqrt{2GM/R}$. We can relate this to the average density. Substituting $M = \rho V = \rho(\frac{4}{3}\pi R^3)$ into the expression for escape speed gives

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{8\pi G\rho}{3}}R \qquad (13.29)$$

Using either form of this equation, you can show that the escape speed for a body at the surface of our sun is $v = 6.18 \times 10^5$ m/s (about 2.2 million km/h, or 1.4 million mi/h). This value, roughly $\frac{1}{500}$ the speed of light, is independent of the mass of the escaping body; it depends on only the mass and radius (or average density and radius) of the sun.

Now consider various stars with the same average density ρ and different radii R. Equation (13.29) shows that for a given value of density ρ, the escape speed v is directly proportional to R. In 1783 the Rev. John Mitchell, an amateur astronomer, noted that if a body with the same average density as the sun had about 500 times the radius of the sun, its escape speed would be greater than the speed of light c. With his statement that "all light emitted from such a body would be made to return toward it," Mitchell became the first person to suggest the existence of what we now call a **black hole**—an object that exerts a gravitational force on other bodies but cannot emit any light of its own.

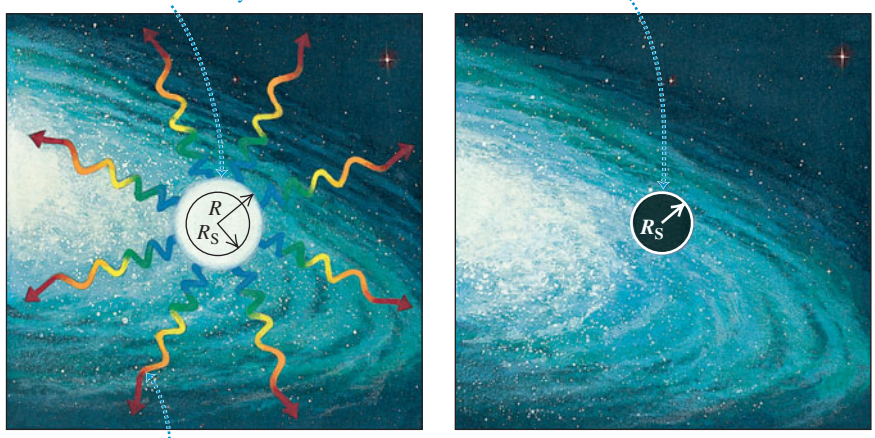
Black Holes, the Schwarzschild Radius, and the Event Horizon

The first expression for escape speed in Eq. (13.29) suggests that a body of mass M will act as a black hole if its radius R is less than or equal to a certain critical radius. How can we determine this critical radius? You might think that you can find the answer by simply setting $v = c$ in Eq. (13.29). As a matter of fact, this does give the correct result, but only because of two compensating errors.

13.26 (a) A body with a radius R greater than the Schwarzschild radius R_S. (b) If the body collapses to a radius smaller than R_S, it is a black hole with an escape speed greater than the speed of light. The surface of the sphere of radius R_S is called the event horizon of the black hole.

(a) When the radius R of a body is greater than the Schwarzschild radius R_S, light can escape from the surface of the body.

(b) If all the mass of the body lies inside radius R_S, the body is a black hole: No light can escape from it.



Gravity acting on the escaping light "red shifts" it to longer wavelengths.

The kinetic energy of light is *not* $mc^2/2$, and the gravitational potential energy near a black hole is *not* given by Eq. (13.9). In 1916, Karl Schwarzschild used Einstein's general theory of relativity (in part a generalization and extension of Newtonian gravitation theory) to derive an expression for the critical radius R_S, now called the **Schwarzschild radius.** The result turns out to be the same as though we had set $v = c$ in Eq. (13.29), so

$$c = \sqrt{\frac{2GM}{R_S}}$$

Solving for the Schwarzschild radius R_S, we find

$$R_S = \frac{2GM}{c^2} \qquad \text{(Schwarzschild radius)} \tag{13.30}$$

If a spherical, nonrotating body with mass M has a radius less than R_S, then *nothing* (not even light) can escape from the surface of the body, and the body is a black hole (Fig. 13.26). In this case, any other body within a distance R_S of the center of the black hole is trapped by the gravitational attraction of the black hole and cannot escape from it.

The surface of the sphere with radius R_S surrounding a black hole is called the **event horizon:** Since light can't escape from within that sphere, we can't see events occurring inside. All that an observer outside the event horizon can know about a black hole is its mass (from its gravitational effects on other bodies), its electric charge (from the electric forces it exerts on other charged bodies), and its angular momentum (because a rotating black hole tends to drag space—and everything in that space—around with it). All other information about the body is irretrievably lost when it collapses inside its event horizon.

Example 13.11 Black hole calculations

Astrophysical theory suggests that a burned-out star whose mass is at least three solar masses will collapse under its own gravity to form a black hole. If it does, what is the radius of its event horizon?

SOLUTION

IDENTIFY, SET UP, and EXECUTE: The radius in question is the Schwarzschild radius. We use Eq. (13.30) with a value of M equal to three solar masses, or $M = 3(1.99 \times 10^{30}\ \text{kg}) = 6.0 \times 10^{30}\ \text{kg}$:

$$R_S = \frac{2GM}{c^2} = \frac{2(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(6.0 \times 10^{30}\ \text{kg})}{(3.00 \times 10^8\ \text{m/s})^2}$$

$$= 8.9 \times 10^3\ \text{m} = 8.9\ \text{km} = 5.5\ \text{mi}$$

EVALUATE: The average density of such an object is

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{6.0 \times 10^{30}\,\text{kg}}{\frac{4}{3}\pi(8.9 \times 10^3\,\text{m})^3} = 2.0 \times 10^{18}\,\text{kg/m}^3$$

This is about 10^{15} times as great as the density of familiar matter on earth and is comparable to the densities of atomic nuclei. In fact, once the body collapses to a radius of R_S, nothing can prevent it from collapsing further. All of the mass ends up being crushed down to a single point called a *singularity* at the center of the event horizon. This point has zero volume and so has *infinite* density.

A Visit to a Black Hole

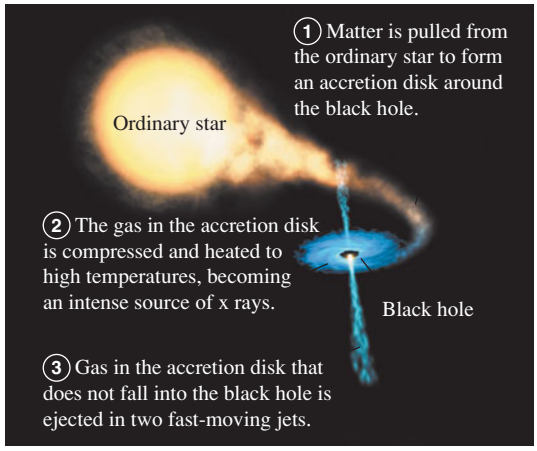
At points far from a black hole, its gravitational effects are the same as those of any normal body with the same mass. If the sun collapsed to form a black hole, the orbits of the planets would be unaffected. But things get dramatically different close to the black hole. If you decided to become a martyr for science and jump into a black hole, the friends you left behind would notice several odd effects as you moved toward the event horizon, most of them associated with effects of general relativity.

If you carried a radio transmitter to send back your comments on what was happening, your friends would have to retune their receiver continuously to lower and lower frequencies, an effect called the *gravitational red shift*. Consistent with this shift, they would observe that your clocks (electronic or biological) would appear to run more and more slowly, an effect called *time dilation*. In fact, during their lifetimes they would never see you make it to the event horizon.

In your frame of reference, you would make it to the event horizon in a rather short time but in a rather disquieting way. As you fell feet first into the black hole, the gravitational pull on your feet would be greater than that on your head, which would be slightly farther away from the black hole. The *differences* in gravitational force on different parts of your body would be great enough to stretch you along the direction toward the black hole and compress you perpendicular to it. These effects (called *tidal forces*) would rip you to atoms, and then rip your atoms apart, before you reached the event horizon.

Detecting Black Holes

If light cannot escape from a black hole and if black holes are as small as Example 13.11 suggests, how can we know that such things exist? The answer is that any gas or dust near the black hole tends to be pulled into an *accretion disk* that swirls around and into the black hole, rather like a whirlpool (Fig. 13.27). Friction within the accretion disk's material causes it to lose mechanical energy



13.27 A binary star system in which an ordinary star and a black hole orbit each other. The black hole itself cannot be seen, but the x rays from its accretion disk can be detected.

and spiral into the black hole; as it moves inward, it is compressed together. This causes heating of the material, just as air compressed in a bicycle pump gets hotter. Temperatures in excess of 10^6 K can occur in the accretion disk, so hot that the disk emits not just visible light (as do bodies that are "red-hot" or "white-hot") but x rays. Astronomers look for these x rays (emitted by the material *before* it crosses the event horizon) to signal the presence of a black hole. Several promising candidates have been found, and astronomers now express considerable confidence in the existence of black holes.

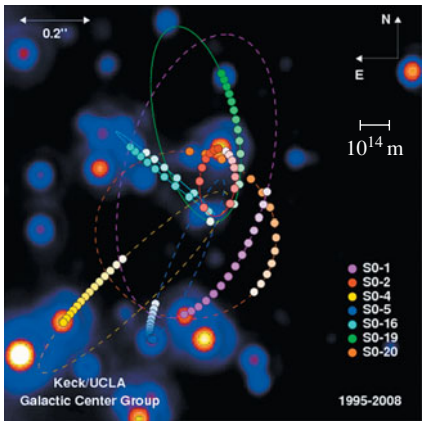
Black holes in binary star systems like the one depicted in Fig. 13.27 have masses a few times greater than the sun's mass. There is also mounting evidence for the existence of much larger *supermassive black holes*. One example is thought to lie at the center of our Milky Way galaxy, some 26,000 light-years from earth in the direction of the constellation Sagittarius. High-resolution images of the galactic center reveal stars moving at speeds greater than 1500 km/s about an unseen object that lies at the position of a source of radio waves called Sgr A* (Fig. 13.28). By analyzing these motions, astronomers can infer the period T and semi-major axis a of each star's orbit. The mass m_X of the unseen object can then be calculated using Kepler's third law in the form given in Eq. (13.17), with the mass of the sun m_S replaced by m_X:

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_X}} \qquad \text{so} \qquad m_X = \frac{4\pi^2 a^3}{GT^2}$$

The conclusion is that the mysterious dark object at the galactic center has a mass of 8.2×10^{36} kg, or 4.1 *million* times the mass of the sun. Yet observations with radio telescopes show that it has a radius no more than 4.4×10^{10} m, about one-third of the distance from the earth to the sun. These observations suggest that this massive, compact object is a black hole with a Schwarzschild radius of 1.1×10^{10} m. Astronomers hope to improve the resolution of their observations so that they can actually see the event horizon of this black hole.

Other lines of research suggest that even larger black holes, in excess of 10^9 times the mass of the sun, lie at the centers of other galaxies. Observational and theoretical studies of black holes of all sizes continue to be an exciting area of research in both physics and astronomy.

13.28 This false-color image shows the motions of stars at the center of our galaxy over a 13-year period. Analyzing these orbits using Kepler's third law indicates that the stars are moving about an unseen object that is some 4.1×10^6 times the mass of the sun. The scale bar indicates a length of 10^{14} m (670 times the distance from the earth to the sun) at the distance of the galactic center.



Test Your Understanding of Section 13.8 If the sun somehow collapsed to form a black hole, what effect would this event have on the orbit of the earth? (i) The orbit would shrink; (ii) the orbit would expand; (iii) the orbit would remain the same size.

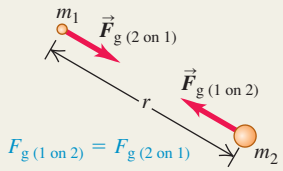


I

CHAPTER 13 SUMMARY

Newton's law of gravitation: *Any* two bodies with masses m_1 and m_2, a distance r apart, attract each other with forces inversely proportional to r^2. These forces form an action–reaction pair and obey Newton's third law. When two or more bodies exert gravitational forces on a particular body, the total gravitational force on that individual body is the vector sum of the forces exerted by the other bodies. The gravitational interaction between spherical mass distributions, such as planets or stars, is the same as if all the mass of each distribution were concentrated at the center. (See Examples 13.1–13.3 and 13.10.)

$$F_g = \frac{Gm_1m_2}{r^2} \qquad (13.1)$$



Gravitational force, weight, and gravitational potential energy: The weight w of a body is the total gravitational force exerted on it by all other bodies in the universe. Near the surface of the earth (mass m_E and radius R_E), the weight is essentially equal to the gravitational force of the earth alone. The gravitational potential energy U of two masses m and m_E separated by a distance r is inversely proportional to r. The potential energy is never positive; it is zero only when the two bodies are infinitely far apart. (See Examples 13.4 and 13.5.)

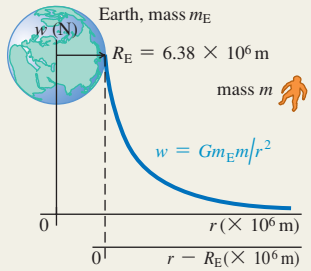
$$w = F_g = \frac{Gm_Em}{R_E^2} \qquad (13.3)$$

(weight at earth's surface)

$$g = \frac{Gm_E}{R_E^2} \qquad (13.4)$$

(acceleration due to gravity at earth's surface)

$$U = -\frac{Gm_Em}{r} \qquad (13.9)$$



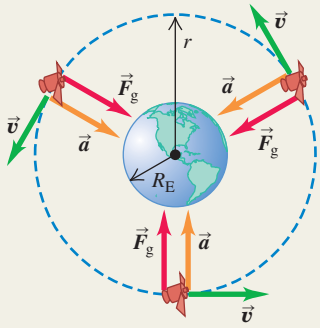
Orbits: When a satellite moves in a circular orbit, the centripetal acceleration is provided by the gravitational attraction of the earth. Kepler's three laws describe the more general case: an elliptical orbit of a planet around the sun or a satellite around a planet. (See Examples 13.6–13.9.)

$$v = \sqrt{\frac{Gm_E}{r}}$$

(speed in circular orbit) (13.10)

$$T = \frac{2\pi r}{v} = 2\pi r\sqrt{\frac{r}{Gm_E}} = \frac{2\pi r^{3/2}}{\sqrt{Gm_E}}$$

(period in circular orbit) (13.12)



Black holes: If a nonrotating spherical mass distribution with total mass M has a radius less than its Schwarzschild radius R_S, it is called a black hole. The gravitational interaction prevents anything, including light, from escaping from within a sphere with radius R_S. (See Example 13.11.)

$$R_S = \frac{2GM}{c^2}$$

(Schwarzschild radius) (13.30)



If all of the body is inside its Schwarzschild radius $R_S = 2GM/c^2$, the body is a black hole.

BRIDGING PROBLEM Speeds in an Elliptical Orbit

A comet orbits the sun (mass m_S) in an elliptical orbit of semi-major axis a and eccentricity e. (a) Find expressions for the speeds of the comet at perihelion and aphelion. (b) Evaluate these expressions for Comet Halley (see Example 13.9).

SOLUTION GUIDE

See MasteringPhysics® study area for a Video Tutor solution. MP

IDENTIFY and SET UP

1. Sketch the situation; show all relevant dimensions. Label the perihelion and aphelion.
2. List the unknown quantities, and identify the target variables.
3. Just as for a satellite orbiting the earth, the mechanical energy is conserved for a comet orbiting the sun. (Why?) What other quantity is conserved as the comet moves in its orbit? (*Hint:* See Section 13.5.)

EXECUTE

4. You'll need at least two equations that involve the two unknown speeds, and you'll need expressions for the sun–comet distances at perihelion and aphelion. (*Hint:* See Fig. 13.18.)
5. Solve the equations for your target variables. Compare your expressions: Which speed is lower? Does this make sense?
6. Use your expressions from step 5 to find the perihelion and aphelion speeds for Comet Halley. (*Hint:* See Appendix F.)

EVALUATE

7. Check whether your results make sense for the special case of a circular orbit ($e = 0$).

Problems

For instructor-assigned homework, go to www.masteringphysics.com MP

•, ••, •••: Problems of increasing difficulty. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q13.1 A student wrote: "The only reason an apple falls downward to meet the earth instead of the earth rising upward to meet the apple is that the earth is much more massive and so exerts a much greater pull." Please comment.

Q13.2 A planet makes a circular orbit with period T around a star. If it were to orbit, at the same distance, a star with three times the mass of the original star, would the new period (in terms of T) be (a) $3T$, (b) $T\sqrt{3}$, (c) T, (d) $T/\sqrt{3}$, or (e) $T/3$?

Q13.3 If all planets had the same average density, how would the acceleration due to gravity at the surface of a planet depend on its radius?

Q13.4 Is a pound of butter on the earth the same amount as a pound of butter on Mars? What about a kilogram of butter? Explain.

Q13.5 Example 13.2 (Section 13.1) shows that the acceleration of each sphere caused by the gravitational force is inversely proportional to the mass of that sphere. So why does the force of gravity give all masses the same acceleration when they are dropped near the surface of the earth?

Q13.6 When will you attract the sun more: today at noon, or tonight at midnight? Explain.

Q13.7 Since the moon is constantly attracted toward the earth by the gravitational interaction, why doesn't it crash into the earth?

Q13.8 A planet makes a circular orbit with period T around a star. If the planet were to orbit at the same distance around this star, but had three times as much mass, what would the new period (in terms of T) be: (a) $3T$, (b) $T\sqrt{3}$, (c) T, (d) $T/\sqrt{3}$, or (e) $T/3$?

Q13.9 The sun pulls on the moon with a force that is more than twice the magnitude of the force with which the earth attracts the moon. Why, then, doesn't the sun take the moon away from the earth?

Q13.10 As defined in Chapter 7, gravitational potential energy is $U = mgy$ and is positive for a body of mass m above the earth's surface (which is at $y = 0$). But in this chapter, gravitational potential energy is $U = -Gm_E m/r$, which is *negative* for a body of mass m above the earth's surface (which is at $r = R_E$). How can you reconcile these seemingly incompatible descriptions of gravitational potential energy?

Q13.11 A planet is moving at constant speed in a circular orbit around a star. In one complete orbit, what is the net amount of work done on the planet by the star's gravitational force: positive, negative, or zero? What if the planet's orbit is an ellipse, so that the speed is not constant? Explain your answers.

Q13.12 Does the escape speed for an object at the earth's surface depend on the direction in which it is launched? Explain. Does your answer depend on whether or not you include the effects of air resistance?

Q13.13 If a projectile is fired straight up from the earth's surface, what would happen if the total mechanical energy (kinetic plus potential) is (a) less than zero, and (b) greater than zero? In each case, ignore air resistance and the gravitational effects of the sun, the moon, and the other planets.

Q13.14 Discuss whether this statement is correct: "In the absence of air resistance, the trajectory of a projectile thrown near the earth's surface is an *ellipse,* not a parabola."

Q13.15 The earth is closer to the sun in November than in May. In which of these months does it move faster in its orbit? Explain why.

Q13.16 A communications firm wants to place a satellite in orbit so that it is always directly above the earth's 45th parallel (latitude 45° north). This means that the plane of the orbit will not pass through the center of the earth. Is such an orbit possible? Why or why not?